

Comprehensive Manual of

BODY FLUID ANALYSIS

Principles, Procedures, Calculations,
Interpretation and Reporting

For Laboratory Professionals and Students

- Synovial Fluid
- Ascitic Fluid
- Pleural Fluid
- Cerebrospinal Fluid (CSF)
- Pericardial Fluid
- Semen Analysis
- Urine Analysis
- Stool Analysis
- Cell Count Procedures



Mr Shankar Shahi

BSc. MLT, MSc. TQM

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Quality Control Expert



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Copyright and Publication Details

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Principles, Procedures, Calculations, Interpretation and Reporting

Author: Mr Shankar Shahi, BSc. MLT, MSc. TQM

Published by Shankar Shahi, Kathmandu, Nepal

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This manual is intended for educational and professional reference use. It does not replace patient-specific clinical judgment, medical diagnosis, treatment decisions, local laboratory policy, instrument manufacturer instructions or validated laboratory procedures.

Reference Intervals and Interpretation

Use the reference intervals, decision limits, methods and critical-result policies validated by the reporting laboratory. Interpret laboratory findings together with specimen quality, clinical information and other investigations. Microscopy illustrations are educational visual references and are not patient-specific diagnostic images.

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Preface

Body fluid examination brings together specimen handling, physical assessment, microscopy, cell counting, biochemical testing and clinical interpretation. This manual presents these activities in a structured format for medical laboratory professionals and students.

The eight chapters cover synovial, ascitic, pleural, cerebrospinal and pericardial fluids, followed by semen, urine and stool analysis. Each chapter brings together practical examination procedures, calculations where applicable, interpretation, quality control and reporting guidance.

My aim is to make the sequence of laboratory work easier to follow: understand the specimen, perform the examination systematically, assess the quality of the result and report the findings clearly. The report formats and workflows can support training and the development of local procedures, with adaptation and approval within the laboratory.

As in Clinical Correlation of Laboratory Findings, this manual encourages readers to consider the pattern of findings and its clinical context. A laboratory finding is one part of the diagnostic process. The limitations and interpretive notes within the chapters are therefore as important as the procedural steps.

I hope this book serves as a useful companion at the laboratory bench and in teaching, while supporting careful observation, reliable reporting and continued learning.

Mr Shankar Shahi
Kathmandu, Nepal
2026

How to Use This Manual

Find the relevant specimen

Use the chapter overview for direct access to the eight specimen chapters. The detailed contents list the numbered sections within each chapter. Lists of tables and figures provide separate navigation to the captioned material.

Follow the examination sequence

Start with specimen requirements and collection, then proceed to physical examination, cell counts, microscopy and additional investigations as described in the chapter.

Read calculations with their context

Check the stated units, dilution, chamber area or volume and the worked example together. Cell count procedures appear within the relevant fluid chapters rather than as a separate ninth chapter.

Use the reporting formats

The specimen-specific report formats are teaching and adaptation resources. Match them to the methods, units, authorization process and documentation requirements of your laboratory.

Check interpretation and limitations

Read the interpretation, quality control, limitations and urgent communication sections alongside the procedure. Use the laboratory's approved policies when applying this material.

Use the page numbers

Preliminary pages use Roman numerals. The main text begins at page 1. The contents refer to printed page numbers; a PDF viewer's file-page counter may be different.

Abbreviations

ADA Adenosine deaminase

AFB Acid-fast bacilli

CPPD Calcium pyrophosphate dihydrate

CSF Cerebrospinal fluid

EDTA Ethylenediaminetetraacetic acid

HPF High-power field

LDH Lactate dehydrogenase

LPF Low-power field

MSU Monosodium urate

PMN Polymorphonuclear neutrophil

QC Quality control

RBC Red blood cell

SAAG Serum–ascites albumin gradient

SBP Spontaneous bacterial peritonitis

SOP Standard operating procedure

UTI Urinary tract infection

WBC White blood cell

WHO World Health Organization

Contents

CHAPTER OVERVIEW

Copyright and Publication Details	ii
Preface.....	iii
How to Use This Manual	iv
Abbreviations	v
Detailed Contents	vii
List of Tables	xv
List of Figures	xvi
Chapter 1. Synovial Fluid Analysis	1
Chapter 2. Ascitic Fluid Analysis.....	13
Chapter 3. Pleural Fluid Analysis.....	31
Chapter 4. Cerebrospinal Fluid (CSF) Analysis.....	52
Chapter 5. Pericardial Fluid Analysis	65
Chapter 6. Semen Analysis.....	75
Chapter 7. Urine Analysis	88
Chapter 8. Stool Analysis.....	106
Appendix A. Microscopy Plates	123
Selected References and Further Reading	126
About the Author	127

Detailed Contents

Chapter 1. Synovial Fluid Analysis	1
1. Introduction	1
2. Purpose	1
3. Scope	1
4. Principle	1
5. Specimen	2
6. Specimen Collection and Distribution	2
7. Specimen Transport and Storage	2
8. Specimen Rejection Criteria	2
9. Required Equipment and Materials	3
10. Laboratory Examination	3
11. Total Cell Count	4
12. RBC Count	4
13. Differential Leukocyte Count	5
14. Crystal Examination	5
15. Monosodium Urate Crystals — Gout	5
16. Calcium Pyrophosphate Dihydrate Crystals — CPPD	6
17. Other Possible Crystals	6
18. Gram Stain	6
19. Culture	7
20. Biochemical Tests	7
21. Interpretation of Synovial Fluid	7
22. Common Clinical Patterns	8
23. Quality Control	9
24. Limitations	9
25. Safety Precautions	9
26. READY-TO-USE SYNOVIAL FLUID REPORT FORMAT	10
27. Suggested Standard Comments	11
28. Suggested Critical/Immediate Communication	11
29. Summary Workflow	11
Chapter 2. Ascitic Fluid Analysis	13
1. Introduction	13
2. Purpose	13
3. Scope	13
4. Principle	14
5. Specimen	14
6. Specimen Collection	14
7. Specimen Transport	14
8. Specimen Rejection / Qualification Criteria	15
9. Required Equipment and Materials	15
10. Laboratory Examination	16
11. Physical Examination	16
12. Color	16
13. Clarity	16
14. Clot Formation	16
15. Total Nucleated Cell / WBC Count	17
16. RBC Count	17
17. Differential Cell Count	17
18. Absolute Neutrophil / PMN Count	18

19. Spontaneous Bacterial Peritonitis — SBP	18
20. Corrected PMN Count in Bloody Ascitic Fluid	18
21. Biochemical Examination	18
22. ASCITIC FLUID ALBUMIN	19
23. SERUM–ASCITES ALBUMIN GRADIENT — SAAG	19
24. SAAG Interpretation	19
25. Ascitic Fluid Total Protein	19
26. Combined SAAG and Protein Interpretation	19
27. Traditional Transudate vs Exudate Classification	20
28. Glucose	20
29. Lactate Dehydrogenase — LDH	20
30. Secondary Bacterial Peritonitis	20
31. Gram Stain	21
32. Culture	21
33. Culture Interpretation	21
34. Amylase	22
35. Triglyceride	22
36. Chylous vs Pseudochylous Fluid	22
37. Ascitic Fluid ADA	22
38. AFB Examination	23
39. Cytology Examination	23
40. Malignant Ascites	23
41. Tuberculous Ascites	24
42. Pancreatic Ascites	24
43. Cardiac Ascites	24
44. Cirrhotic Ascites	24
45. General Interpretation Table	24
46. Quality Control	25
47. Limitations	25
48. Safety Precautions	26
49. READY-TO-USE ASCITIC FLUID REPORT FORMAT	27
50. Suggested Standard Laboratory Comments	28
51. Critical / Urgent Result Communication	29
52. Practical Laboratory Workflow	29
53. Quick Interpretation Guide	30
54. Conclusion	30
Chapter 3. Pleural Fluid Analysis	31
1. Introduction	31
2. Purpose	31
3. Scope	31
4. Principle	32
5. Specimen	32
6. Specimen Collection and Distribution	32
7. Paired Serum Samples	32
8. Specimen for Pleural Fluid pH	33
9. Specimen Transport	33
10. Specimen Rejection / Qualification Criteria	33
11. Required Equipment and Materials	33
12. Laboratory Examination	34
13. Physical Examination	34
14. Color and Appearance	34
15. Appearance and Possible Interpretation	34
16. Clot Formation	35

17. Total Nucleated Cell / WBC Count.....	35
18. Manual WBC Count.....	35
19. RBC Count.....	35
20. Differential Cell Count.....	35
21. Differential Cell Pattern	36
22. Routine Biochemical Examination.....	36
23. Pleural Fluid Total Protein	36
24. Pleural Fluid LDH.....	36
25. LIGHT'S CRITERIA	37
26. Light's Criteria Calculation – Example	37
27. Transudative Effusion	37
28. Exudative Effusion.....	38
29. Light's Criteria and Heart Failure.....	38
30. Pleural Fluid Glucose.....	38
31. Pleural Fluid pH	39
32. Complicated Parapneumonic Effusion	39
33. Empyema	39
34. Gram Stain	40
35. Culture.....	40
36. Tuberculous Pleural Effusion.....	40
37. Pleural Fluid ADA	41
38. AFB / Tuberculosis Testing	41
39. Malignant Pleural Effusion	41
40. Cytological Examination.....	41
41. Cytology Reporting.....	42
42. Chylothorax.....	42
43. Chylothorax Interpretation	42
44. Causes of Chylothorax	43
45. Pseudochylothorax	43
46. Hemothorax.....	43
47. Example Hemothorax Calculation	43
48. Pleural Fluid Amylase	43
49. General Interpretation Table	44
50. Quality Control	44
51. Limitations	45
52. Safety Precautions.....	45
53. READY-TO-USE PLEURAL FLUID REPORT FORMAT	46
54. Suggested Standard Laboratory Comments	47
55. Critical / Urgent Result Communication.....	48
56. Practical Laboratory Workflow.....	48
57. Quick Interpretation Guide	50
58. Key Laboratory Points	50
59. Conclusion	51
Chapter 4. Cerebrospinal Fluid (CSF) Analysis.....	52
1. Introduction	52
2. Purpose.....	52
3. Specimen.....	52
4. Specimen Collection	53
5. Specimen Transport	53
6. Specimen Rejection / Qualification.....	53
7. Required Equipment and Materials.....	53
8. Laboratory Examination.....	54
9. Physical Examination.....	54

10. Normal Appearance	54
11. Xanthochromia.....	54
12. Total WBC Count	55
13. Manual WBC Count.....	55
14. RBC Count.....	55
15. Traumatic Tap and WBC Correction	55
16. Differential Cell Count.....	56
17. Differential Interpretation	56
18. CSF Protein.....	56
19. CSF Glucose	57
20. CSF / Serum Glucose Ratio	57
21. Typical CSF Patterns.....	57
22. Bacterial Meningitis	57
23. Viral Meningitis	58
24. Tuberculous Meningitis	58
25. Fungal / Cryptococcal Meningitis	58
26. Gram Stain	59
27. Bacterial Culture	59
28. Molecular Testing	59
29. CSF Lactate.....	59
30. Cytology.....	60
31. Other Specialized Investigations	60
32. Quality Control	60
33. Safety	61
34. READY-TO-USE CSF REPORT FORMAT.....	62
35. Suggested Standard Comments.....	63
36. Urgent Communication.....	63
37. Practical Workflow	63
38. Quick Interpretation	64
39. Conclusion	64
Chapter 5. Pericardial Fluid Analysis.....	65
1. Introduction.....	65
2. Purpose.....	65
3. Specimen.....	65
4. Clinical Importance.....	66
5. Specimen Distribution.....	66
6. Transport	66
7. Physical Examination.....	66
8. Appearance.....	66
9. Cell Count	67
10. Differential Count	67
11. Biochemical Examination	67
12. Important Limitation – Exudate/Transudate Classification.....	67
13. Protein	68
14. LDH	68
15. Glucose	68
16. Purulent Pericarditis	68
17. Gram Stain	68
18. Bacterial Culture	69
19. Tuberculous Pericarditis.....	69
20. ADA	69
21. Cytology.....	69
22. Cytology Procedure.....	69

23. Cytology Reporting.....	70
24. Flow Cytometry	70
25. Hemorrhagic Pericardial Effusion.....	70
26. Chylous Pericardial Effusion	70
27. General Interpretation Table	71
28. Quality Control	71
29. Safety	71
30. READY-TO-USE PERICARDIAL FLUID REPORT FORMAT.....	72
31. Mandatory Interpretation Note.....	73
32. Suggested Comments	73
33. Urgent Communication.....	73
34. Practical Workflow	73
35. Conclusion	74
Chapter 6. Semen Analysis.....	75
1. Introduction.....	75
2. Purpose.....	75
3. Important Interpretation Principle.....	75
4. Patient Preparation	76
5. Patient Instructions.....	76
6. Collection Method.....	76
7. Transport	76
8. Specimen Reception.....	76
9. Macroscopic Examination.....	77
10. Liquefaction	77
11. Appearance.....	77
12. Semen Volume.....	77
13. Viscosity	77
14. pH.....	78
15. Microscopic Examination	78
16. Sperm Concentration.....	78
17. Counting Method	78
18. Total Sperm Number.....	79
19. Sperm Motility	79
20. Motility Calculation	79
21. WHO 2021 Motility Distribution.....	79
22. Sperm Vitality.....	79
23. Vitality Principle	80
24. Morphology.....	80
25. Normal Morphology.....	80
26. Morphological Abnormalities	80
27. Agglutination	81
28. Round Cells.....	81
29. Leukocytes	81
30. Semen Culture.....	81
31. WHO 2021 Reference Distribution – Quick Table.....	81
32. Common Semen Terminology	82
33. Azoospermia Verification	82
34. Cryptozoospermia	82
35. Total Motile Sperm Number – Optional Calculation.....	82
36. Quality Control	83
37. Factors Affecting Results	83
38. Repeat Testing.....	83
39. READY-TO-USE SEMEN ANALYSIS REPORT FORMAT.....	84

40. Example Interpretation Patterns	85
41. Practical Workflow	85
42. Key Laboratory Points	86
43. Conclusion	87
Chapter 7. Urine Analysis	88
1. Introduction	88
2. Purpose	88
3. Scope	88
4. Specimen	89
5. Types of Urine Specimens	89
6. Collection Instructions	90
7. Transport and Storage	90
8. Specimen Rejection / Qualification Criteria	90
9. Required Equipment and Materials	91
10. Laboratory Examination	91
11. Physical Examination	91
12. Color	91
13. Appearance / Clarity	91
14. Specific Gravity	92
15. Chemical Examination	92
16. Urine pH	92
17. Protein	93
18. Causes of Proteinuria	93
19. Glucose	94
20. Ketones	94
21. Blood	94
22. Bilirubin	94
23. Urobilinogen	95
24. Nitrite	95
25. Leukocyte Esterase	95
26. Microscopic Examination	95
27. Low-Power Examination	96
28. High-Power Examination	96
29. Red Blood Cells	96
30. RBC Morphology	96
31. White Blood Cells	96
32. Causes of Pyuria	97
33. Epithelial Cells	97
34. Bacteria	97
35. Yeast	98
36. Casts	98
37. Cast Interpretation	98
38. Crystals	99
39. Common Crystals	99
40. Important Abnormal Crystals	99
41. Calcium Oxalate	100
42. Triple Phosphate	100
43. Mucus	100
44. Spermatozoa	100
45. Trichomonas	100
46. Typical UTI Pattern	101
47. Typical Glomerular Pattern	101
48. Typical Tubular Injury Pattern	101

49. Quality Control	101
50. Microscopy QC	101
51. Limitations	102
52. READY-TO-USE URINE REPORT FORMAT	103
53. Suggested Standard Comments.....	104
54. Practical Workflow	104
55. Quick Interpretation Guide	104
56. Conclusion	105
Chapter 8. Stool Analysis.....	106
1. Introduction.....	106
2. Purpose.....	106
3. Scope.....	106
4. Specimen.....	106
5. Patient Preparation	107
6. Sample Collection	107
7. Transport	107
8. Preserved Stool	108
9. Multiple Stool Samples for Parasites	108
10. Specimen Rejection / Qualification.....	108
11. Required Equipment and Materials.....	108
12. Laboratory Examination.....	109
13. Macroscopic Examination.....	109
14. Consistency.....	109
15. Color	110
16. Mucus.....	110
17. Visible Blood	110
18. Worms / Proglottids	110
19. Direct Microscopic Examination	111
20. Saline Wet Mount	111
21. Saline Preparation	111
22. Iodine Wet Mount	111
23. Iodine Preparation	112
24. Microscopic Elements	112
25. RBC in Stool	112
26. WBC / Pus Cells	112
27. Protozoal Parasites	112
28. Entamoeba histolytica	113
29. Important Entamoeba Reporting Limitation	113
30. Giardia duodenalis	113
31. Helminth Eggs and Larvae.....	114
32. Ascaris lumbricoides.....	114
33. Hookworm	114
34. Trichuris trichiura	114
35. Strongyloides stercoralis	114
36. Taenia Species.....	115
37. Concentration Technique	115
38. Formalin–Ethyl Acetate Sedimentation	115
39. Permanent Staining	115
40. Modified Acid-Fast Stain	115
41. Cryptosporidium	116
42. Parasite Antigen Testing	116
43. Stool Culture	116
44. Cholera / Vibrio Investigation.....	116

46. Molecular Gastrointestinal Panels.....	116
47. Occult Blood	117
48. Fecal Fat.....	117
49. Stool pH / Reducing Substances	117
50. Quality Control	117
53. Limitations	118
54. READY-TO-USE STOOL REPORT FORMAT	119
55. Suggested Standard Reports.....	120
56. Urgent / Important Communication	120
57. Practical Laboratory Workflow.....	120
58. Quick Interpretation Guide	121
59. Key Laboratory Points	122
60. Conclusion	122
Appendix A. Microscopy Plates	123

List of Tables

Table 1.1. Specimen Collection and Distribution	2
Table 1.2. Synovial Fluid Color Interpretation	3
Table 1.3. Interpretation of Synovial Fluid	7
Table 2.1. Ascitic Fluid Specimen Collection.....	14
Table 2.2. Ascitic Fluid Color and Appearance Interpretation.....	16
Table 2.3. Combined SAAG and Protein Interpretation	19
Table 2.4. General Interpretation of Ascitic Fluid	24
Table 3.1. Pleural Fluid Specimen Collection and Distribution.....	32
Table 3.2. Pleural Fluid Appearance and Possible Interpretation	34
Table 3.3. General Interpretation of Pleural Fluid	44
Table 4.1. CSF Specimen Collection	53
Table 4.2. CSF Normal and Abnormal Appearance.....	54
Table 4.3. CSF Differential Cell Count.....	56
Table 4.4. Typical CSF Patterns.....	57
Table 5.1. Pericardial Fluid Specimen Distribution	66
Table 5.2. Pericardial Fluid Appearance and Possible Association	66
Table 5.3. Pericardial Fluid Differential Count.....	67
Table 5.4. General Interpretation of Pericardial Fluid	71
Table 6.1. WHO 2021 Reference Distribution – Quick Table	81
Table 7.1. Urine Chemical Examination.....	92
Table 7.2. Urine Cast Interpretation	98

List of Figures

Figure 1.1. Synovial Fluid laboratory workflow	11
Figure 2.1. Ascitic Fluid laboratory workflow	29
Figure 3.1. Pleural Fluid laboratory workflow	48
Figure 4.1. CSF laboratory workflow	63
Figure 5.1. Pericardial Fluid laboratory workflow	73
Figure 6.1. Semen laboratory workflow	85
Figure 7.1. Urine laboratory workflow	104
Figure 8.1. Stool laboratory workflow	120
Figure A.1. Stool parasite microscopy plate	123
Figure A.2. Urine sediment microscopy plate	124
Figure A.3. Body fluid microscopy plate	125

The three microscopy plates supplied at the end of the manuscript are grouped in Appendix A.

Chapter 1. Synovial Fluid Analysis

Laboratory Procedure and Reporting Manual

1. Introduction

Synovial fluid is a clear, viscous fluid present within synovial joints. It is derived mainly from plasma ultrafiltrate and contains hyaluronic acid produced by synovial cells. Its major functions are lubrication of the joint surfaces, reduction of friction, and provision of nutrients to articular cartilage.

Analysis of synovial fluid is particularly useful in differentiating:

- Non-inflammatory joint disease
- Inflammatory arthritis
- Crystal-induced arthritis
- Septic/infectious arthritis
- Hemorrhagic or traumatic joint effusion

Synovial fluid examination is especially important in acute monoarthritis because infection and crystal-associated arthritis may present similarly. Detection of crystals does **not** exclude concomitant septic arthritis.

2. Purpose

To establish a standardized procedure for collection, processing, examination, interpretation, quality control, and reporting of synovial fluid in the clinical laboratory.

3. Scope

This procedure applies to laboratory personnel performing:

- Physical examination
- Total nucleated cell/WBC count
- RBC count when required
- Differential leukocyte count
- Crystal identification
- Gram stain
- Microbiological culture
- Selected biochemical tests
- Synovial fluid reporting

4. Principle

Joint diseases cause changes in the physical, cellular, biochemical, crystalline, and microbiological characteristics of synovial fluid.

Laboratory analysis assists in classifying the fluid into:

- Normal
- Non-inflammatory
- Inflammatory
- Septic/infectious
- Hemorrhagic

Identification of monosodium urate and calcium pyrophosphate crystals using compensated polarized light microscopy is particularly useful for diagnosing gout and calcium pyrophosphate deposition disease.

5. Specimen

Specimen: Synovial fluid obtained by arthrocentesis.

Common sites include:

- Knee, Ankle, Wrist, Elbow, Shoulder, Hip, Small joints of hands and feet

The specimen must be collected by an appropriately trained clinician using aseptic technique.

6. Specimen Collection and Distribution

Whenever sufficient volume is available, divide the sample according to the investigations required.

Table 1.1. Specimen Collection and Distribution

Container	Recommended Examination
Sterile plain container	Gram stain, aerobic/anaerobic culture
EDTA tube	Cell count and differential
EDTA or validated heparin tube	Crystal examination
Plain/red-top tube	Chemistry such as glucose/protein if requested
Cytology container	Cytological examination when indicated

For cell count and crystal analysis, EDTA or heparin-containing specimens are accepted by major reference laboratories. Samples should be delivered to the laboratory promptly.

A recommended specimen volume may be approximately **3–5 mL per container when available**, but small-volume samples should not automatically be rejected because microscopic examination may still be possible from only a few drops.

Priority when sample volume is limited

When only a small volume is obtained and septic arthritis is clinically suspected, prioritize:

Microbiology → Cell count/differential → Crystal examination → Other tests

7. Specimen Transport and Storage

Synovial fluid should preferably be transported to the laboratory immediately and examined as soon as possible.

For crystal analysis, prompt examination at room temperature is preferable because changes in temperature and pH may affect crystal formation or solubility.

Do not freeze routine synovial fluid specimens.

Delayed samples should be handled according to the laboratory's validated stability procedure.

8. Specimen Rejection Criteria

Consider rejection or qualification of the sample for:

1. Unlabeled or incorrectly labeled specimen
2. Leaking container
3. Wrong specimen source
4. Gross contamination
5. Inappropriate container for requested test
6. Clotted specimen when accurate cell count is required
7. Excessively delayed specimen without acceptable storage
8. Frozen specimen where freezing is not validated
9. Insufficient specimen volume for requested investigations

Where the sample is clinically irreplaceable, consultation with the clinician is preferable to automatic rejection.

9. Required Equipment and Materials

1. Personal protective equipment
2. Test tubes and disposable pipettes
3. Glass slides and coverslips
4. Light microscope
5. Polarizing microscope
6. Red compensator/first-order red plate where available
7. Improved Neubauer counting chamber
8. Cell-counting diluent
9. Centrifuge
10. Cytocentrifuge, if available
11. Wright, Wright-Giemsa, or May-Grünwald-Giemsa stain
12. Gram stain reagents
13. Culture media
14. Automated body-fluid-compatible cell counter, if validated
15. Timer
16. Applicator sticks
17. Sterile loops
18. Quality-control materials where applicable

10. Laboratory Examination

A. Physical Examination

Perform physical examination before centrifugation.

10.1 Volume

Record the actual volume received.

Example: **Volume received: 4.0 mL**

Do not interpret volume without considering the joint from which the specimen was obtained.

10.2 Color

- **Report as:**
 - Colorless, Pale yellow, Yellow, Dark yellow, Yellow-green, Pink, Red, Brown
- **Interpretation**

Table 1.2. Synovial Fluid Color Interpretation

Appearance	Possible Significance
Clear/colorless	Normal
Pale yellow	Non-inflammatory
Yellow	Inflammatory
Yellow-green	Possible infection
Pink/red	Hemorrhage or traumatic tap
Brown	Old hemorrhage or degraded blood

10.3 Clarity

Report as: Clear, slightly hazy, Hazy, Cloudy, Turbid, Opaque, Purulent

Normal synovial fluid is clear. Increasing numbers of leukocytes, RBCs, crystals, microorganisms, fibrin, and debris increase turbidity.

10.4 Viscosity

Normal synovial fluid has high viscosity because of hyaluronic acid. A simple qualitative assessment may be performed by allowing fluid to drop slowly from a syringe or applicator.

Report as: High/normal, moderately decreased, Low, very low

Inflammatory fluid commonly shows reduced viscosity.

10.5 Clot Formation

Observe for spontaneous clot formation.

Report as: None, Small clot, Moderate clot, large clot

Fibrin clot formation may occur in inflammatory conditions because increased synovial permeability permits fibrinogen to enter the joint fluid.

11. Total Cell Count

11.1 Manual WBC Count

Principle

Cells present in a known volume of synovial fluid are counted using a hemocytometer, and the result is expressed as cells/ μ L.

Procedure

1. Mix specimen gently.
2. Prepare an appropriate dilution if necessary.
3. Load both sides of an Improved Neubauer chamber.
4. Allow cells to settle.
5. Count nucleated cells according to the laboratory's validated chamber-counting protocol.
6. Calculate the result.

General calculation

$$\text{Cells}/\mu\text{L} = \text{Number of cells counted} \times \text{Dilution factor} \times 10 \div \text{Number of large squares counted}$$

Example:

If 120 cells are counted in 4 large squares using a 1:20 dilution:

$$\text{WBC} = 120 \times 20 \times 10 \div 4$$

$$= 6,000 \text{ cells}/\mu\text{L}$$

Highly viscous samples may require pretreatment according to the laboratory's validated procedure.

12. RBC Count

RBC count may be performed manually using a hemocytometer or by a validated automated analyzer.

Report as:

RBC: _____ cells/ μ L

High RBC counts may indicate:

- Traumatic aspiration
- Trauma
- Hemarthrosis
- Bleeding disorder
- Anticoagulant-associated bleeding
- Intra-articular injury

Interpret in conjunction with specimen appearance and clinical findings.

13. Differential Leukocyte Count

Procedure

- Mix the specimen.
- Prepare direct smears or preferably cytocentrifuge preparation when cell concentration is low.
- Air-dry.
- Stain using Wright-Giemsa or equivalent stain.
- Examine microscopically.
- Count approximately 100 nucleated cells when sufficient cells are available.
- Classify cells.

Report:

- Neutrophils %
- Lymphocytes %
- Monocytes/macrophages %
- Eosinophils %, if present
- Other cells, if present

Typical interpretation

Predominance of neutrophils suggests an acute inflammatory process and may occur in septic arthritis and crystal-induced arthritis. A high neutrophil proportion therefore should not be interpreted independently of culture and crystal results.

14. Crystal Examination

Crystal examination should preferably be performed on fresh, well-mixed synovial fluid using ordinary and compensated polarized light microscopy.

Compensated polarized light microscopy remains the standard method for identifying monosodium urate and calcium pyrophosphate crystals.

Wet Preparation

1. Mix specimen gently.
2. Place one drop of synovial fluid on a clean glass slide.
3. Apply coverslip.
4. Examine initially under ordinary light.
5. Examine under polarized light.
6. Apply red compensator if available.
7. Search both extracellular and intracellular areas.
8. Record crystal type, morphology and birefringence.

15. Monosodium Urate Crystals — Gout

Morphology

1. Needle-shaped
2. Slender
3. May be intracellular or extracellular

Polarization

Strong negative birefringence

With a red compensator:

- Yellow when parallel to the compensator slow axis
- Blue when perpendicular

Report

Monosodium urate crystals: Present

or

Needle-shaped, strongly negatively birefringent crystals consistent with monosodium urate identified.

Finding monosodium urate crystals strongly supports gout. (Merck Manuals)

16. Calcium Pyrophosphate Dihydrate Crystals — CPPD

Commonly associated with pseudogout/CPPD disease.

Morphology

1. Rhomboid
2. Rectangular
3. Short rod-shaped

Polarization

Weak positive birefringence

With a red compensator:

- Blue when parallel
- Yellow when perpendicular

Report

Calcium pyrophosphate crystals: Present

or

Rhomboid-shaped, weakly positively birefringent crystals consistent with calcium pyrophosphate identified.

17. Other Possible Crystals

Occasionally the following may be encountered:

- Cholesterol crystals
- Calcium oxalate crystals
- Basic calcium phosphate/hydroxyapatite

Hydroxyapatite crystals are extremely small and generally cannot be reliably identified using routine polarized light microscopy.

Artifacts should always be considered.

18. Gram Stain

When infection is suspected:

1. Mix specimen.
2. Prepare smear directly or from centrifuged sediment depending on cellularity.
3. Air-dry.
4. Fix appropriately.
5. Perform Gram staining.
6. Examine systematically under oil immersion.

Report:

- No organisms seen

or

- Gram-positive cocci seen

- Gram-negative bacilli seen
- Gram-negative diplococci seen
- Other morphology as appropriate

Also report approximate leukocyte response where relevant.

Example:

Numerous polymorphonuclear leukocytes seen. No organisms seen.

A negative Gram stain does **not** exclude septic arthritis.

19. Culture

Depending on clinical indications, perform:

- Aerobic bacterial culture
- Anaerobic culture
- Fungal culture
- Mycobacterial culture
- Molecular testing/PCR where available

Synovial fluid from suspected septic arthritis should routinely be considered for Gram stain and aerobic/anaerobic culture.

20. Biochemical Tests

Biochemistry may include selected tests such as:

- Glucose
- Protein
- Lactate
- Other specialized tests

For glucose interpretation, simultaneous serum glucose may be useful.

However, synovial fluid glucose and protein have limited sensitivity and specificity for diagnosing septic arthritis and should not replace cell count, differential, Gram stain and culture.

21. Interpretation of Synovial Fluid

The following ranges are **general interpretive guides rather than absolute diagnostic cutoffs**.

Table 1.3. Interpretation of Synovial Fluid

Parameter	Normal	Non-inflammatory	Inflammatory	Septic/Infectious
Color	Clear	Straw/pale yellow	Yellow	Yellow-green
Clarity	Transparent	Clear/translucent	Cloudy	Turbid/opaque
Viscosity	High	High	Decreased	Very low
WBC / μ L	<200	200–2,000	2,000–100,000	Often >50,000
PMN %	<25%	<25%	Often >50%	Commonly >75%
Crystals	Absent	Absent	May be present	Usually absent*
Gram stain	Negative	Negative	Negative	May be positive
Culture	Negative	Negative	Negative	May be positive

These values overlap considerably. A synovial fluid WBC below **50,000 cells/ μ L** does **not** exclude septic arthritis, particularly in early, partially treated, atypical, gonococcal or prosthetic-joint infection.

*Crystals and infection can occur simultaneously.

22. Common Clinical Patterns

Osteoarthritis

Typical findings:

- Clear or slightly cloudy
- High viscosity
- WBC usually $<2,000/\mu\text{L}$
- Low neutrophil proportion
- No crystals unless another condition coexists

Rheumatoid/Inflammatory Arthritis

Typical findings:

- Yellow/cloudy
- Reduced viscosity
- Increased WBC
- Neutrophilic predominance possible
- Culture usually negative

Gout

Typical findings:

- Cloudy inflammatory fluid
- Increased WBC
- Neutrophil predominance
- Needle-shaped negatively birefringent MSU crystals

CPPD/Pseudogout

Typical findings:

- Inflammatory fluid
- Increased WBC
- Rhomboid, weak positively birefringent CPP crystals

Septic Arthritis

Typical findings:

- Turbid or purulent fluid
- Low viscosity
- Markedly increased WBC
- Predominantly neutrophils
- Gram stain may be positive
- Culture may be positive

A WBC $>50,000$ cells/ μL increases suspicion for infection, particularly with strong neutrophil predominance, but no single cell-count cutoff can independently confirm or exclude septic arthritis. (Merck Manuals)

23. Quality Control

Cell Count

1. Ensure counting chamber is clean and correctly loaded.
2. Avoid bubbles and overfilling.
3. Count both chambers where required.
4. Repeat counts when chamber-to-chamber variation exceeds laboratory limits.
5. Verify automated counts in abnormal or flagged specimens according to analyzer policy.

Differential Count

1. Use properly prepared and stained smears.
2. Review abnormal or unidentified cells by experienced personnel.
3. Use cytopsin when appropriate.

Crystal Examination

1. Maintain polarizing microscope alignment.
2. Use known positive teaching/control slides periodically.
3. Ensure personnel competency in distinguishing MSU and CPP crystals.
4. Avoid contamination and crystalline artifacts.

Microbiology

Use routine microbiological internal QC according to laboratory policy.

24. Limitations

1. Cell-count ranges overlap between inflammatory and septic conditions.
2. Prior antibiotic treatment may reduce culture positivity.
3. Crystals do not exclude simultaneous infection.
4. Improper anticoagulant or contamination can create crystal-like artifacts.
5. Delayed examination can affect cells and biochemical constituents.
6. Small numbers of crystals may be missed.
7. CPP crystals are generally less strongly birefringent than urate crystals and can be more difficult to identify.
8. Automated analyzers must be specifically validated for synovial/body-fluid analysis.

25. Safety Precautions

All synovial fluid specimens should be considered potentially infectious.

Use:

- Gloves
- Laboratory coat/gown
- Eye/face protection where splash risk exists
- Biological safety cabinet for aerosol-generating procedures where appropriate

Dispose of contaminated materials according to biomedical-waste policy. Never recap contaminated needles.

26. READY-TO-USE SYNOVIAL FLUID REPORT FORMAT

LABORATORY NAME: _____			
SYNOVIAL FLUID ANALYSIS REPORT			
PATIENT / SPECIMEN INFORMATION			
Patient Name	_____	Age/Sex	_____
Patient ID	_____	Joint / Site	_____
Date/Time Collected	_____	Date/Time Received	_____
Clinical Information	_____	Specimen	Synovial Fluid
PHYSICAL EXAMINATION			
Parameter	Result	Unit	
Volume Received	_____	mL	
Color	_____		
Appearance / Clarity	_____		
Viscosity	_____		
Clot	_____		
CELL COUNT			
Parameter	Result	Unit	
Total Nucleated Cell / WBC	_____	cells/ μ L	
RBC	_____	cells/ μ L	
DIFFERENTIAL COUNT			
Cell Type	Result		
Neutrophils	_____ %		
Lymphocytes	_____ %		
Monocytes / Macrophages	_____ %		
Eosinophils	_____ %		
Others	_____ %		
CRYSTAL EXAMINATION			
Monosodium urate crystals	☐ Not Seen ☐ Present		
Calcium pyrophosphate crystals	☐ Not Seen ☐ Present		
Other crystals	_____		
Morphology / birefringence	_____		
MICROBIOLOGY			
Gram Stain	_____		
Aerobic Culture	_____		
Anaerobic Culture	_____		
Other Culture / PCR	_____		
BIOCHEMISTRY, IF REQUESTED			
Test	Result	Unit	
Glucose	_____	mg/dL	
Protein	_____	g/dL	
Other	_____		
INTERPRETATION / COMMENTS			

LABORATORY NAME: _____			
Reported By _____	Checked By _____	Authorized By _____	Date/Time _____

27. Suggested Standard Comments

When MSU crystals are detected

“Needle-shaped, strongly negatively birefringent crystals consistent with monosodium urate are identified.”

When CPP crystals are detected

“Rhomboid-shaped, weakly positively birefringent crystals consistent with calcium pyrophosphate are identified.”

When no crystals are found

“No monosodium urate or calcium pyrophosphate crystals identified in the examined preparation.”

When septic arthritis is suspected

“Marked neutrophilic leukocytosis is present. Findings should be correlated with Gram stain, culture and clinical features to exclude septic arthritis.”

Important report comment

“Presence of crystals does not exclude concomitant septic arthritis. Clinical and microbiological correlation is recommended.”

28. Suggested Critical/Immediate Communication

The laboratory should consider immediate communication to the responsible clinician according to its critical-result policy when there is:

- Positive Gram stain from synovial fluid
- Obvious microorganisms
- Strong laboratory evidence suggesting septic arthritis
- Unexpected malignant/abnormal cells
- Other findings designated critical by institutional policy

Document:

Result: _____ **communicated to:** _____
Date: _____ **Time:** _____ **Communicated by:** _____

29. Summary Workflow

Figure 1.1. Synovial Fluid laboratory workflow

Synovial Fluid Received

- Check identification and specimen condition
- Record volume and appearance
- Allocate sterile sample for microbiology
- Perform cell count
- Prepare differential smear
- Perform crystal examination
- Perform Gram stain/culture when indicated
- Perform requested chemistry
- Review all results together
- Interpret/classify findings
- Authorize report
- Immediately communicate critical microbiological findings

Conclusion

Synovial fluid examination is a valuable diagnostic tool for differentiating non-inflammatory, inflammatory, crystal-associated and infectious joint disorders. Reliable interpretation requires integration of physical findings, cell count, differential count, crystal examination, Gram stain, culture and clinical information rather than reliance on any single parameter.

Chapter 2. Ascitic Fluid Analysis

Laboratory Procedure, Interpretation and Reporting Manual

1. Introduction

Ascitic fluid is the abnormal accumulation of fluid within the peritoneal cavity. Small amounts of physiologic peritoneal fluid may normally be present; however, excessive accumulation is termed **ascites**.

Ascites may occur due to several disorders, most commonly:

- Liver cirrhosis and portal hypertension
- Congestive heart failure
- Malignancy
- Tuberculous peritonitis
- Bacterial peritonitis
- Pancreatitis
- Nephrotic syndrome
- Budd–Chiari syndrome
- Lymphatic obstruction
- Chylous ascites
- Bowel perforation or secondary bacterial peritonitis

Laboratory examination of ascitic fluid helps determine the underlying cause of fluid accumulation and, importantly, detect infection such as **spontaneous bacterial peritonitis (SBP)**.

2. Purpose

To provide a standardized laboratory procedure for:

- Collection and handling of ascitic fluid
- Physical examination
- Cell count and differential
- Biochemical analysis
- Serum–Ascites Albumin Gradient (SAAG)
- Microbiological examination
- Cytological examination
- Special investigations
- Interpretation
- Reporting of ascitic fluid

3. Scope

This procedure applies to all laboratory personnel performing analysis of ascitic/peritoneal fluid.

Routine examination may include:

- Physical appearance
- Total nucleated cell count/WBC
- RBC count
- Differential cell count
- Ascitic fluid albumin
- Total protein
- SAAG calculation
- Gram stain
- Bacterial culture

Additional tests are performed depending upon clinical indication.

4. Principle

Ascitic fluid changes in response to different pathological processes.

Laboratory evaluation is based on:

- Gross appearance
- Number and type of cells
- Albumin concentration
- Total protein concentration
- Serum–Ascites Albumin Gradient
- Microorganisms
- Biochemical constituents
- Presence of malignant cells
- Special markers such as triglyceride, amylase and ADA

The **Serum–Ascites Albumin Gradient (SAAG)** is preferred over the traditional transudate/exudate classification for determining whether ascites is associated with portal hypertension.

5. Specimen

Specimen: Ascitic / Peritoneal Fluid

The fluid is obtained by **abdominal paracentesis** performed by an appropriately trained clinician using aseptic technique.

6. Specimen Collection

Collect the specimen into appropriate containers according to the investigations requested.

Table 2.1. Ascitic Fluid Specimen Collection

Container	Investigation
EDTA tube	Cell count and differential
Sterile plain container	Gram stain and microbiology
Blood culture bottles	Aerobic and anaerobic culture
Plain/serum-type tube	Albumin, total protein, glucose, LDH, amylase, triglyceride etc.
Sterile container	AFB testing/TB culture when indicated
Cytology container	Cytological examination

Important Collection Requirement for SAAG

A **serum albumin** sample should preferably be collected on the **same day and near the time of paracentesis**.

Both serum albumin and ascitic fluid albumin must be measured using compatible methods.

7. Specimen Transport

1. Transport the specimen to the laboratory as soon as possible.
2. Avoid unnecessary delay.
3. Cell counts should preferably be performed promptly because cells may deteriorate during prolonged storage.
4. Microbiology specimens should be processed without unnecessary delay.
5. Blood culture bottles should be inoculated appropriately when infection is suspected.

When spontaneous bacterial peritonitis is suspected, bedside inoculation of adequate ascitic fluid into aerobic and anaerobic blood culture bottles improves organism recovery.

8. Specimen Rejection / Qualification Criteria

Consider rejection or qualification in the following situations:

- Unlabeled specimen
- Incorrect patient identification
- Leaking container
- Gross external contamination
- Wrong container for requested test
- Clotted EDTA specimen for cell count
- Insufficient specimen volume
- Excessive delay without appropriate storage
- Sample collected in an inappropriate anticoagulant
- Missing specimen source

Because ascitic fluid may be difficult to recollect, clinically important specimens should not be discarded without consultation with the clinician whenever possible.

9. Required Equipment and Materials

1. Personal protective equipment
2. Test tubes
3. Disposable pipettes
4. Glass slides
5. Coverslips
6. Light microscope
7. Improved Neubauer counting chamber
8. Automated body-fluid cell counter, if validated
9. Cytocentrifuge, if available
10. Centrifuge
11. Wright-Giemsa / May-Grünwald-Giemsa stain
12. Gram stain reagents
13. Culture media
14. Blood culture bottles
15. Biochemistry analyzer
16. Albumin reagent
17. Total protein reagent
18. Glucose reagent
19. LDH reagent
20. Amylase reagent
21. Triglyceride reagent
22. ADA testing system where available

10. Laboratory Examination

Ascitic fluid examination can be divided into:

- A. Physical examination
- B. Cell count
- C. Differential count

- D. Biochemical examination
- E. Microbiological examination
- F. Cytological examination
- G. Special investigations

11. Physical Examination

Physical appearance should be recorded before centrifugation.

11.1 Volume

Record the amount of specimen received.

Example:

Volume received: 20 mL

The volume received in the laboratory does not necessarily represent the total amount of ascites present in the patient.

12. Color

Report as appropriate:

Colorless, Pale yellow, Straw colored, Yellow, Cloudy yellow, Greenish, Pink, Red, Brown, Milky white

General Interpretation

Table 2.2. Ascitic Fluid Color and Appearance Interpretation

Appearance	Possible Association
Clear/straw colored	Cirrhosis/portal hypertension
Cloudy/turbid	Increased cells, infection or malignancy
Purulent	Severe bacterial infection
Bloody/red	Traumatic tap, malignancy, hemorrhage
Milky	Chylous ascites
Dark brown/bile stained	Possible biliary leakage
Greenish	Biliary or gastrointestinal process may be considered

Appearance alone should never be used for definitive diagnosis.

13. Clarity

Report as:

- Clear, slightly hazy, Hazy, Cloudy, Turbid, Opaque, Purulent

Turbidity may result from:

- WBCs, RBCs, Microorganisms, Malignant cells, Lipid/chyle, Cellular debris

14. Clot Formation

Report:

- No clot, small clot, Moderate clot, large clot

Clot formation may occur when fibrinogen enters the peritoneal fluid during inflammatory conditions.

15. Total Nucleated Cell / WBC Count

WBC may be measured using:

- Manual hemocytometer
- Automated body-fluid analyzer validated for ascitic fluid

15.1 Manual Cell Count

Procedure

- Mix the specimen gently but thoroughly.
- Prepare appropriate dilution if required.
- Load the Improved Neubauer counting chamber.
- Allow cells to settle.
- Count nucleated cells according to the laboratory's validated procedure.
- Calculate cells per microliter.

• General Formula

$$\text{Cells}/\mu\text{L} = \text{Cells counted} \times \text{Dilution factor} \times 10 \div \text{Number of large squares counted}$$

16. RBC Count

Report RBC as:

RBC: _____ cells/ μL

Elevated RBC may be associated with:

- Traumatic paracentesis
- Malignancy
- Hemorrhagic ascites
- Abdominal trauma
- Pancreatitis
- Recent surgery or procedure

Grossly bloody fluid should be interpreted carefully.

17. Differential Cell Count

Prepare:

- Direct smear, or preferably
- Cytocentrifuge preparation when available

Stain using Wright-Giemsa, May-Grünwald-Giemsa or another validated stain.

Count approximately **100 nucleated cells** when sufficient cells are available.

Report:

- **Neutrophils %**
- **Lymphocytes %**
- **Monocytes/macrophages %**
- **Mesothelial cells %**
- **Eosinophils %**
- **Other cells**

18. Absolute Neutrophil / PMN Count

For evaluation of spontaneous bacterial peritonitis, the **absolute PMN count** is more important than the total WBC count alone.

Calculation

$$\text{Absolute PMN count} = \text{Total nucleated cells} \times \text{Neutrophil \%} \div 100$$

Example

$$\text{Total nucleated cells} = 1,200 \text{ cells}/\mu\text{L}$$

$$\text{Neutrophils} = 60\%$$

$$\text{Absolute PMN} = 1,200 \times 60 \div 100$$

$$= 720 \text{ cells}/\mu\text{L}$$

19. Spontaneous Bacterial Peritonitis — SBP

An ascitic fluid:

$$\text{PMN count} \geq 250 \text{ cells}/\mu\text{L}$$

is strongly suggestive of **spontaneous bacterial peritonitis in the appropriate clinical setting**, particularly in patients with cirrhosis when no surgically treatable intra-abdominal source of infection is identified.

A positive culture is **not required** for the laboratory finding of neutrocytic ascites.

20. Corrected PMN Count in Bloody Ascitic Fluid

Blood contamination may falsely increase the ascitic PMN count.

For markedly bloody fluid, a correction may be applied:

Corrected PMN

$$\text{Corrected PMN} = \text{Measured PMN} - (\text{RBC count} \div 250)$$

In other words:

Subtract approximately 1 PMN for every 250 RBCs/ μL .

Example

$$\text{Measured PMN} = 400 \text{ cells}/\mu\text{L}$$

$$\text{RBC} = 25,000 \text{ cells}/\mu\text{L}$$

Correction:

$$25,000 \div 250 = 100$$

Corrected PMN:

$$400 - 100$$

$$= 300 \text{ cells}/\mu\text{L}$$

Interpret correction together with clinical findings and institutional policy.

21. Biochemical Examination

Routine biochemical investigation generally includes:

- Albumin
- Total protein

Additional testing depending on clinical suspicion may include:

- Glucose, LDH, Amylase, Triglyceride, Bilirubin, ADA, Other specialized tests

22. ASCITIC FLUID ALBUMIN

Ascitic fluid albumin should be measured together with serum albumin when determining the cause of ascites.

Report:

Ascitic Fluid Albumin: _____ g/dL

23. SERUM–ASCITES ALBUMIN GRADIENT — SAAG

SAAG is one of the most useful measurements for classifying ascites.

Formula

$$\text{SAAG} = \text{Serum Albumin} - \text{Ascitic Fluid Albumin}$$

Both results should preferably be from specimens obtained on the same day.

Example

Serum albumin = 3.2 g/dL

Ascitic fluid albumin = 1.0 g/dL

SAAG: $3.2 - 1.0 = 2.2$ g/dL

24. SAAG Interpretation**SAAG ≥ 1.1 g/dL**

Suggests ascites associated with **portal hypertension**.

Common causes include:

- Liver cirrhosis
- Alcohol-associated hepatitis with portal hypertension
- Congestive heart failure
- Budd–Chiari syndrome
- Portal vein/vascular disorders producing portal hypertension

SAAG < 1.1 g/dL

Suggests ascites **not primarily caused by portal hypertension**.

Possible causes include:

- Peritoneal malignancy
- Tuberculous peritonitis
- Pancreatic ascites
- Serositis
- Certain infections
- Nephrotic syndrome in some settings

SAAG should always be interpreted with clinical findings and other fluid investigations.

25. Ascitic Fluid Total Protein

Report: Total Protein: _____ g/dL

Total protein is particularly useful when interpreted together with SAAG.

26. Combined SAAG and Protein Interpretation

Table 2.3. Combined SAAG and Protein Interpretation

SAAG	Ascitic Protein	Possible Association
≥ 1.1 g/dL	< 2.5 g/dL	Cirrhosis commonly
≥ 1.1 g/dL	≥ 2.5 g/dL	Cardiac ascites, Budd–Chiari etc.
< 1.1 g/dL	Usually, high	Malignancy, TB, pancreatic/peritoneal disease

This table is an interpretive guide rather than an absolute diagnostic classification.

27. Traditional Transudate vs Exudate Classification

Historically, ascitic fluid was divided into:

Transudate

Often associated with:

- Cirrhosis
- Portal hypertension
- Cardiac causes

Exudate

Often associated with:

- Infection
- Malignancy
- Tuberculosis
- Pancreatitis

However, **SAAG is generally preferred** because total protein alone may misclassify ascites.

28. Glucose

Ascitic fluid glucose may be measured when infection, malignancy or secondary peritonitis is suspected.

Low glucose may occur with:

- Bacterial infection
- Tuberculous peritonitis
- Malignancy
- Secondary bacterial peritonitis

Very low ascitic glucose, particularly ≤ 50 mg/dL, may raise suspicion for secondary bacterial peritonitis when associated with other abnormalities.

29. Lactate Dehydrogenase — LDH

Ascitic LDH may be useful when differentiating uncomplicated portal hypertensive ascites from inflammatory or secondary bacterial processes.

Markedly increased LDH may occur with:

- Infection
- Malignancy
- Secondary bacterial peritonitis

Ascitic LDH greater than the laboratory serum upper reference limit may be a clue to secondary bacterial peritonitis.

30. Secondary Bacterial Peritonitis

Secondary bacterial peritonitis may occur due to:

- Gastrointestinal perforation
- Abscess
- Bowel ischemia
- Inflamed or perforated abdominal organ

It is clinically important to distinguish secondary bacterial peritonitis from spontaneous bacterial peritonitis.

Findings raising suspicion include:

- Very high PMN count
- Polymicrobial Gram stain
- Polymicrobial culture
- Ascitic total protein > 1.0 g/dL
- Glucose ≤ 50 mg/dL
- Ascitic LDH above the serum upper reference limit
- Failure of PMN count to decrease with appropriate treatment

A commonly used biochemical approach is the presence of at least **two of the following three findings**:

- Total protein > 1.0 g/dL
- Glucose < 50 mg/dL
- LDH above the upper reference limit for serum

Such findings should prompt urgent clinical assessment for a secondary intra-abdominal source.

31. Gram Stain

Procedure

- Mix specimen.
- Centrifuge an appropriate volume when necessary.
- Prepare smear from sediment.
- Air dry.
- Fix appropriately.
- Perform Gram staining.
- Examine under oil immersion.

Report:

- No organisms seen or Gram-positive cocci seen, Gram-negative bacilli seen etc.

The presence of numerous neutrophils should also be reported when appropriate.

Example:

Numerous polymorphonuclear leukocytes seen. No organisms seen.

A negative Gram stain does not exclude bacterial peritonitis.

32. Culture

When SBP is suspected, culture should ideally be performed before antibiotic treatment.

Recommended microbiological investigation may include:

- Aerobic culture
- Anaerobic culture

Bedside inoculation of ascitic fluid into blood culture bottles substantially improves organism recovery.

33. Culture Interpretation

PMN $\geq$ 250 cells/ μ L + Positive Culture

Consistent with **culture-positive spontaneous bacterial peritonitis** in the appropriate clinical context if there is no secondary intra-abdominal source.

PMN $\geq$ 250 cells/ μ L + Negative Culture

Often termed: **Culture-Negative Neutrocytic Ascites**

This finding remains clinically important and may represent SBP despite negative culture.

PMN $<$ 250 cells/ μ L + Positive Culture

May be termed:

Non-neutrocytic Bacterascites

Clinical correlation and follow-up are required.

34. Amylase

Ascitic fluid amylase may be requested when pancreatic disease or gastrointestinal perforation is suspected.

Markedly increased ascitic fluid amylase may occur in:

- Pancreatic ascites
- Pancreatic pseudocyst leakage
- Acute/chronic pancreatitis
- Gastrointestinal perforation

Results should be compared with serum amylase and the clinical picture.

35. Triglyceride

Ascitic triglyceride should be measured when fluid appears:

- Milky, Creamy, White, Chylous

Ascitic triglyceride >200 mg/dL

strongly supports **chylous ascites**.

Chylous ascites may occur with:

- Lymphoma, Other malignancies, Lymphatic obstruction, Abdominal surgery, Trauma, Cirrhosis
- Tuberculosis
- Congenital lymphatic disorders

When available, demonstration of chylomicrons provides definitive biochemical evidence of chyle.

36. Chylous vs Pseudochylous Fluid

Milky fluid should not automatically be called chylous ascites.

Turbidity may also result from:

- Large numbers of leukocytes
- Malignant cells
- Cellular degeneration
- Infection

Therefore, triglyceride measurement is recommended.

37. Ascitic Fluid ADA

Adenosine deaminase — ADA — may be useful when **tuberculous peritonitis** is suspected.

Elevated ADA supports TB in an appropriate clinical context.

However:

- Interpretation depends on assay
- Cutoffs vary by laboratory and population
- Cirrhosis may affect diagnostic performance
- ADA alone should not be used to confirm or exclude tuberculosis

Results should be correlated with:

- AFB stain
- Mycobacterial culture
- Molecular testing where available
- Imaging
- Peritoneal biopsy when clinically indicated

38. AFB Examination

When tuberculosis is suspected:

- Concentrate specimen if appropriate
- Perform AFB smear
- Send for mycobacterial culture
- Perform molecular tests according to institutional protocol

A negative AFB smear does not exclude tuberculous peritonitis.

39. Cytology Examination

Ascitic fluid cytology is especially useful when malignancy is suspected.

Procedure

- Record appearance.
- Mix specimen.
- Centrifuge or prepare cytospin.
- Prepare slides.
- Stain appropriately.

Examine for:

- Mesothelial cells
- Inflammatory cells
- Macrophages
- Atypical cells
- Malignant cells

Report according to cytology laboratory policy.

40. Malignant Ascites

Possible laboratory findings include:

- Bloody fluid
- Increased protein
- SAAG often <1.1 g/dL in peritoneal carcinomatosis
- Increased LDH
- Abnormal/malignant cells on cytology

However, SAAG may be high in malignancy when portal hypertension or extensive liver involvement is present.

Therefore, cytology and clinical correlation are important.

41. Tuberculous Ascites

Common laboratory pattern may include:

- Clear to cloudy/yellow fluid
- Increased nucleated cells
- Lymphocyte predominance
- Increased protein
- SAAG often <1.1 g/dL
- Elevated ADA
- AFB testing may be positive
- Mycobacterial culture may be positive

No single routine laboratory parameter is diagnostic by itself.

42. Pancreatic Ascites

Possible laboratory pattern:

- Increased total protein
- SAAG commonly <1.1 g/dL
- Markedly elevated ascitic amylase
- Pancreatic disease clinically or radiologically present

43. Cardiac Ascites

Typical pattern:

SAAG ≥ 1.1 g/dL

With **Ascitic total protein ≥ 2.5 g/dL** may suggest cardiac ascites in the appropriate clinical setting. Clinical cardiac evaluation remains essential.

44. Cirrhotic Ascites

Typical uncomplicated cirrhotic ascites may demonstrate:

1. Clear/straw-colored fluid
2. SAAG ≥ 1.1 g/dL
3. Total protein often < 2.5 g/dL
4. PMN < 250 cells/ μ L
5. Negative culture

PMN ≥ 250 cells/ μ L should raise concern for SBP.

45. General Interpretation Table

Table 2.4. General Interpretation of Ascitic Fluid

Condition	SAAG	Protein	Predominant Cells / Special Findings
Cirrhosis	≥ 1.1	Usually, < 2.5	Low PMN unless infected
Cardiac ascites	≥ 1.1	Usually, ≥ 2.5	Variable
Budd–Chiari	Usually, ≥ 1.1	Often high	Variable
Peritoneal malignancy	Usually, < 1.1	Usually, high	Malignant cells may be present
Tuberculous peritonitis	Usually, < 1.1	High	Lymphocyte predominance
Pancreatic ascites	Usually, < 1.1	High	Very high amylase
Chylous ascites	Variable	Variable	TG usually > 200 mg/dL
SBP	Variable	Variable	PMN $\geq 250/\mu$ L
Secondary peritonitis	Variable	Often > 1.0	Very high PMN, possible polymicrobial infection

Values overlap and must not be interpreted in isolation.

46. Quality Control

Cell Count

1. Mix sample adequately.
2. Use clean counting chambers.
3. Avoid overfilling or underfilling.
4. Avoid air bubbles.
5. Perform duplicate chamber counts according to laboratory policy.
6. Verify abnormal automated results microscopically where appropriate.
7. Use only analyzers validated for body-fluid specimens.

Differential Count

1. Prepare good-quality smears.
2. Use appropriate stain.
3. Count adequate cells.
4. Have abnormal cells reviewed by experienced personnel.

Biochemistry

1. Perform internal QC according to analyzer policy.

2. Ensure assays are validated or verified for body fluids where required.
3. Use appropriate calibration.
4. Follow reagent manufacturer instructions.

Microbiology

Use appropriate positive and negative controls according to microbiology laboratory policy.

47. Limitations

1. SAAG must be interpreted using serum and ascitic albumin obtained at approximately the same time.
2. Ascitic total protein alone should not be used to classify the cause of ascites.
3. Antibiotic therapy before collection may reduce culture yield.
4. Negative Gram stain does not exclude SBP.
5. Negative bacterial culture does not exclude SBP.
6. PMN counts can be falsely elevated by blood contamination.
7. Cytology may be negative despite underlying malignancy.
8. AFB smear has limited sensitivity in tuberculous peritonitis.
9. ADA results depend on assay and patient population.
10. No single ascitic fluid result should be interpreted without clinical correlation.

48. Safety Precautions

Ascitic fluid should be considered potentially infectious.

Always follow standard precautions.

Use:

- Gloves
- Laboratory coat/gown
- Eye protection where splash risk exists
- Biological safety cabinet for aerosol-generating procedures when appropriate

Dispose of specimens and contaminated materials according to biomedical waste policy.

49. READY-TO-USE ASCITIC FLUID REPORT FORMAT

LABORATORY NAME: _____			
ASCITIC / PERITONEAL FLUID ANALYSIS REPORT			
PATIENT / SPECIMEN INFORMATION			
Patient Name	_____	Age/Sex	_____
Patient ID	_____	Ward / OPD	_____
Clinical Diagnosis	_____	Specimen	Ascitic / Peritoneal Fluid
Date/Time Collected	_____	Date/Time Received	_____
PHYSICAL EXAMINATION			
Parameter	Result	Unit	
Volume Received	_____	mL	
Color	_____		
Appearance / Clarity	_____		
Clot	_____		
CELL COUNT			
Parameter	Result	Unit	
Total Nucleated Cells / WBC	_____	cells/ μ L	
RBC	_____	cells/ μ L	
DIFFERENTIAL COUNT			
Cell Type	Result		
Neutrophils	_____ %		
Lymphocytes	_____ %		
Monocytes / Macrophages	_____ %		
Mesothelial Cells	_____ %		
Eosinophils	_____ %		
Others	_____ %		
Absolute PMN Count	Total cells $\times$ Neutrophil % $\div$ 100 = _____ cells/ μ L		
Corrected PMN, if applicable	_____ cells/ μ L		
BIOCHEMICAL EXAMINATION			
Test	Result	Unit	
Ascitic Fluid Albumin	_____	g/dL	
Serum Albumin	_____	g/dL	
Total Protein	_____	g/dL	
Glucose	_____	mg/dL	
LDH	_____	U/L	
Amylase	_____	U/L	
Triglyceride	_____	mg/dL	
ADA	_____	U/L	
Other	_____		
SERUM-ASCITES ALBUMIN GRADIENT (SAAG)			
Formula	SAAG = Serum Albumin – Ascitic Fluid Albumin		

LABORATORY NAME: _____			
Calculated SAAG	_____ g/dL		
Interpretation	☐ SAAG ≥ 1.1 g/dL — Consistent with portal hypertensive mechanism ☐ SAAG < 1.1 g/dL — Suggests non-portal hypertensive mechanism		
MICROBIOLOGY			
Gram Stain	_____		
Aerobic Culture	_____		
Anaerobic Culture	_____		
AFB Stain	_____		
Mycobacterial Culture / Molecular Test	_____		
CYTOLOGY			
Result	☐ Negative for malignant cells ☐ Atypical cells present ☐ Malignant cells identified ☐ Other: _____		
INTERPRETATION / COMMENT			
Reported By	Checked By	Authorized By	Date/Time
_____	_____	_____	_____

50. Suggested Standard Laboratory Comments

Normal/Uncomplicated Portal Hypertensive Pattern

“Ascitic fluid findings show a high serum–ascites albumin gradient (SAAG ≥ 1.1 g/dL), supporting a portal hypertensive mechanism. Clinical correlation is recommended.”

Low SAAG

“Serum–ascites albumin gradient is < 1.1 g/dL, suggesting a non-portal hypertensive mechanism. Correlation with clinical findings and investigations for peritoneal malignancy, infection, tuberculosis, pancreatic disease or other causes is recommended.”

SBP Suspicion

“Ascitic fluid absolute neutrophil count is ≥ 250 cells/ μ L. Findings are concerning for spontaneous bacterial peritonitis in the appropriate clinical setting. Immediate clinical and microbiological correlation is advised.”

High PMN with Negative Gram Stain

“Marked neutrophilic response is present. No organisms are seen on Gram stain. A negative Gram stain does not exclude bacterial peritonitis. Culture correlation is recommended.”

Chylous Ascites

“Milky ascitic fluid with elevated triglyceride concentration is consistent with chylous ascites. Clinical evaluation for underlying lymphatic obstruction, malignancy, cirrhosis, infection or other causes is recommended.”

TB Pattern

“Lymphocyte-predominant ascitic fluid with a low SAAG/high protein pattern may be seen in tuberculous peritonitis. Correlation with ADA, mycobacterial studies and clinical findings is advised.”

Malignancy Suspicion

“Findings may be associated with malignant ascites. Cytological and clinical correlation is recommended.”

51. Critical / Urgent Result Communication

The laboratory should establish its own critical-result policy.

Findings that may require immediate communication include:

Ascitic fluid PMN ≥ 250 cells/ μL in a patient at risk for SBP

1. Positive Gram stain
2. Organisms detected in ascitic fluid
3. Positive culture from a normally sterile peritoneal specimen
4. Findings strongly suggestive of secondary bacterial peritonitis
5. Unexpected malignant cells, according to institutional policy

Document:

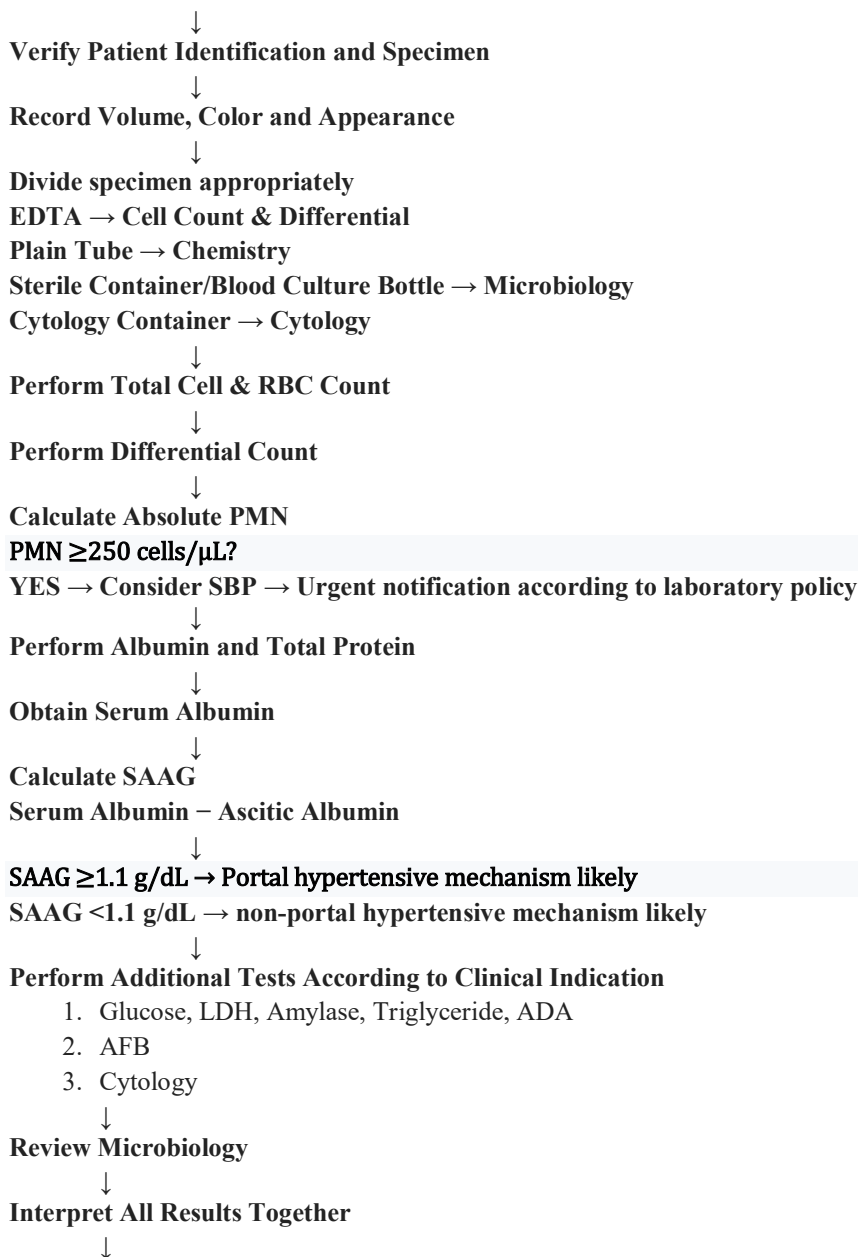
Critical result: _____ **Communicated to:** _____

Date: _____ **Time:** _____

Communicated by: _____

52. Practical Laboratory Workflow

Figure 2.1. Ascitic Fluid laboratory workflow

ASCITIC FLUID RECEIVED

AUTHORIZE FINAL REPORT

53. Quick Interpretation Guide

PMN ≥ 250 cells/ μ L

→ Consider **Spontaneous Bacterial Peritonitis**

SAAG ≥ 1.1 g/dL

→ Consider **Portal Hypertension**

SAAG ≥ 1.1 + Protein < 2.5 g/dL

→ **Cirrhosis pattern commonly**

SAAG ≥ 1.1 + Protein ≥ 2.5 g/dL

→ Consider **Cardiac ascites / vascular causes**

SAAG < 1.1 + High Protein

→ Consider **TB / Malignancy / Pancreatic or other peritoneal disease**

Milky Fluid + Triglyceride > 200 mg/dL

→ **Chylous Ascites**

Very High Amylase

→ Consider **Pancreatic Ascites**

Lymphocyte Predominance + High Protein + Low SAAG

→ Consider **Tuberculous Peritonitis**

Malignant Cells Present

→ **Malignant Ascites**

Multiple Bacterial Species + Low Glucose + High LDH

→ Consider **Secondary Bacterial Peritonitis**

54. Conclusion

Ascitic fluid analysis is an important diagnostic laboratory investigation for determining the etiology of ascites and detecting potentially life-threatening infection.

For routine laboratory evaluation, the most important initial investigations are:

Physical Examination + Cell Count + Differential Count + Albumin + Total Protein + SAAG + Microbiological Examination

The two most important practical calculations are:

SAAG = Serum Albumin – Ascitic Fluid Albumin

and

Absolute PMN = Total Nucleated Cells $\times$ Neutrophil % $\div 100$

An **SAAG ≥ 1.1 g/dL** supports a portal hypertensive mechanism, while an **ascitic PMN count ≥ 250 cells/ μ L** is a key laboratory threshold for spontaneous bacterial peritonitis in the appropriate clinical context.

All findings should be interpreted together with clinical, imaging, microbiological and pathological information.

Chapter 3. Pleural Fluid Analysis

Laboratory Procedure, Interpretation and Reporting Manual

1. Introduction

Pleural fluid is present in a very small quantity within the pleural space between the visceral and parietal pleura. It acts as a lubricant during respiratory movement.

Abnormal accumulation of fluid in the pleural cavity is called a **pleural effusion**.

Common causes include:

- Congestive heart failure
- Liver cirrhosis
- renal disease
- Pneumonia
- Pleural infection/empyema
- Tuberculosis
- Malignancy
- Pulmonary embolism
- Pancreatitis
- Connective-tissue disorders
- Chylothorax
- Hemothorax

Laboratory evaluation of pleural fluid is primarily aimed at distinguishing **transudative from exudative effusions** and identifying specific causes such as infection, tuberculosis, malignancy, chylothorax and hemothorax.

2. Purpose

To establish a standardized laboratory procedure for:

- Specimen collection and handling
- Physical examination
- Total cell count
- Differential leukocyte count
- Biochemical examination
- Light's criteria calculation
- Pleural fluid pH
- Microbiological examination
- Cytology
- Tuberculosis investigations
- Chylothorax investigation
- Hemothorax assessment

3. Scope

This procedure applies to laboratory personnel performing routine and specialized pleural fluid investigations.

Routine examination may include:

- Physical appearance
- Total nucleated cell/WBC count
- RBC count
- Differential count
- Protein
- LDH
- Glucose
- pH when indicated
- Gram stain
- Bacterial culture
- Cytology

Additional tests are selected according to clinical suspicion.

4. Principle

Pleural effusion occurs when there is an imbalance between production and removal of pleural fluid.

Pleural fluid is traditionally classified into:

Transudative Effusion

Usually caused by systemic factors affecting hydrostatic or oncotic pressure.

Common examples:

- Congestive heart failure
- Cirrhosis
- Hypoalbuminemia
- Nephrotic syndrome

Exudative Effusion

Usually caused by local pleural inflammation, infection, malignancy or tissue injury.

Common examples:

- Pneumonia
- Empyema
- Tuberculosis
- Malignancy
- Pulmonary embolism
- Pancreatitis
- Rheumatoid pleuritis

Light's criteria are the standard laboratory method for differentiating exudative from transudative pleural effusions.

5. Specimen

Specimen: Pleural fluid

Pleural fluid is obtained by **thoracentesis/pleural aspiration**, preferably under image guidance. Current British Thoracic Society guidance recommends image-guided thoracentesis to reduce complications. (Thorax)

6. Specimen Collection and Distribution

Whenever sufficient fluid is available, distribute into appropriate containers.

Table 3.1. Pleural Fluid Specimen Collection and Distribution

Container	Investigation
EDTA tube	Cell count and differential
Plain sterile container	Gram stain and microbiology
Aerobic blood culture bottle	Aerobic culture
Anaerobic blood culture bottle	Anaerobic culture
Plain tube	Protein, LDH, glucose, amylase, triglyceride, cholesterol, ADA
Blood gas syringe	Pleural fluid pH
Cytology container	Cytological examination
Sterile container	AFB, TB culture/molecular tests

For suspected malignant pleural effusion, approximately **25–50 mL** should ideally be submitted for initial cytological examination when available. (Thorax)

7. Paired Serum Samples

For classification using Light's criteria, obtain serum samples at approximately the same time for:

- Serum total protein
- Serum LDH

These values are required to calculate:

- Pleural fluid/serum protein ratio
- Pleural fluid/serum LDH ratio

Serum protein and LDH should ideally be measured concurrently with pleural fluid analysis.

8. Specimen for Pleural Fluid pH

Pleural pH requires special handling.

Recommended procedure

1. Collect directly into a blood gas syringe.
2. Avoid contamination with local anesthetic.
3. Expel unnecessary liquid heparin.
4. Avoid air exposure.
5. Cap the syringe immediately.
6. Send rapidly for blood gas analysis.
7. Analyze as soon as possible.

Local anesthetic and excessive heparin can falsely lower pleural pH, while residual air or prolonged delay may raise the measured pH. (Thorax)

9. Specimen Transport

1. Send pleural fluid immediately to the laboratory.
2. Perform cell count promptly.
3. Process microbiology specimens without unnecessary delay.
4. Process pH specimens urgently.
5. When pleural infection is suspected, inoculation into aerobic and anaerobic blood culture bottles improves microbiological yield. (Thorax)

10. Specimen Rejection / Qualification Criteria

Consider rejection or qualification for:

- Unlabeled specimen
- Patient identification mismatch
- Leaking container
- Gross contamination
- Incorrect container
- Clotted specimen intended for cell count
- Excessive delay
- Improperly stored sample
- Insufficient volume

Clinically irreplaceable pleural fluid should not automatically be discarded without consultation with the responsible clinician.

11. Required Equipment and Materials

1. PPE
2. Disposable pipettes
3. Test tubes
4. Glass slides
5. Coverslips
6. Light microscope
7. Improved Neubauer counting chamber
8. Validated automated body-fluid analyzer
9. Cytocentrifuge where available
10. Centrifuge
11. Wright-Giemsa stain
12. Gram stain reagents
13. Microbiological culture media
14. Blood culture bottles

15. Chemistry analyzer
16. Blood gas analyzer
17. Cytology stains and materials

12. Laboratory Examination

Pleural fluid examination can be divided into:

- Physical examination
- Cell count
- Differential count
- Biochemical examination
- Classification by Light's criteria
- Microbiology
- Cytology
- Special investigations

13. Physical Examination

Physical appearance should be documented before centrifugation.

13.1 Volume

Record the actual volume received.

Example: **Volume Received: 30 mL**

This does not represent the total amount of fluid present in the pleural cavity.

14. Color and Appearance

Report as:

- | | | | |
|--------------|---------------|-----------------|------------|
| • Clear | • Pale yellow | • Straw colored | • Yellow |
| • Cloudy | • Turbid | • Purulent | • Pink |
| • Red/bloody | • Milky | • Brown | • Greenish |

15. Appearance and Possible Interpretation

Table 3.2. Pleural Fluid Appearance and Possible Interpretation

Appearance	Possible Association
Clear/straw colored	Transudative effusion, but nonspecific
Cloudy	Increased cells, infection, malignancy
Turbid/purulent	Empyema/pleural infection
Bloody	Trauma, malignancy, pulmonary embolism, hemothorax
Milky	Chylothorax or pseudochylothorax
Foul smelling	Possible anaerobic infection
Food particles	Consider esophageal perforation
Brown	Old hemorrhage or other unusual processes

16. Clot Formation

Report: • No clot • Small clot • Moderate clot • Large clot

Fibrin clot formation may occur in inflammatory or exudative effusions.

17. Total Nucleated Cell / WBC Count

Cell count may be performed using:

- Manual hemocytometer
- Validated automated body-fluid analyzer

18. Manual WBC Count

Procedure

- Mix the specimen gently.
- Dilute if required.
- Load an Improved Neubauer chamber.
- Allow cells to settle.
- Count nucleated cells using the validated counting pattern.
- Calculate the result.

Calculation

$$\text{Cells}/\mu\text{L} = \text{Cells counted} \times \text{Dilution factor} \times 10 \div \text{Number of large squares counted}$$

Report: Total Nucleated Cells/WBC: _____ cells/ μL

19. RBC Count

Report: RBC: _____ cells/ μL

An increased RBC count may be associated with:

- Traumatic thoracentesis
- Hemothorax
- Malignancy
- Pulmonary embolism
- Trauma
- Postoperative effusion

Very bloody pleural fluid should prompt consideration of pleural fluid hematocrit.

20. Differential Cell Count

Prepare:

Direct smear, or Cytocentrifuge preparation where available

Stain with Wright-Giemsa, May-Grünwald-Giemsa or an equivalent validated stain.

Count approximately 100 nucleated cells when possible.

Report:

- Neutrophils %
- Lymphocytes %
- Monocytes/macrophages %
- Eosinophils %
- Mesothelial cells %
- Other cells %

21. Differential Cell Pattern

Neutrophil-Predominant Effusion

Commonly associated with:

- Acute parapneumonic effusion
- Pleural infection
- Pulmonary embolism
- Early inflammatory processes

- Lymphocyte-Predominant Effusion

May be associated with:

- Tuberculosis
- Malignancy
- Lymphoma
- Chronic pleural effusion
- Rheumatoid pleuritis
- Sarcoidosis
- Some chronic heart-failure effusions
- Cellular predominance guides further investigation but is not diagnostic by itself. (AAFP)

22. Routine Biochemical Examination

Routine pleural fluid biochemistry should generally include:

- Total protein
- LDH

Depending on the clinical condition:

- Glucose
- pH
- Albumin
- Amylase
- Triglyceride
- Cholesterol
- ADA

23. Pleural Fluid Total Protein

Report:

Pleural Fluid Protein: _____ g/dL

A paired serum protein is required for Light's criteria.

24. Pleural Fluid LDH

Report:

Pleural Fluid LDH: _____ U/L

A paired serum LDH should be measured.

Also note the laboratory's serum LDH upper reference limit.

25. LIGHT'S CRITERIA

Pleural fluid is classified as **EXUDATIVE** if **one or more** of the following criteria are met:

- Criterion 1
- Pleural Fluid Protein / Serum Protein >0.5
- Criterion 2
- Pleural Fluid LDH / Serum LDH >0.6
- Criterion 3
- Pleural Fluid LDH > two-thirds of the upper reference limit of serum LDH
- If none of the three criteria are met, the fluid is generally classified as transudative. (AAFP)

26. Light's Criteria Calculation – Example

Results

Pleural protein = 4.0 g/dL

Serum protein = 6.5 g/dL

Pleural LDH = 350 U/L

Serum LDH = 400 U/L

Serum LDH upper reference limit = 250 U/L

Protein Ratio

$$4.0 \div 6.5 = 0.62$$

Criterion positive because:

$$0.62 > 0.5$$

LDH Ratio

$$350 \div 400 = 0.875$$

Criterion positive because:

$$0.875 > 0.6$$

Two-thirds Serum LDH Upper Limit

$$250 \times 2/3 = 167 \text{ U/L}$$

Pleural LDH = 350 U/L

Criterion positive because:

- $350 > 167 \text{ U/L}$
- Interpretation
- Exudative Pleural Effusion

27. Transudative Effusion

Common causes include:

- Congestive heart failure
- Cirrhosis
- Hypoalbuminemia
- Nephrotic syndrome
- Some renal disorders
- Hepatic hydrothorax

Typical characteristics:

- Clear/straw-colored fluid
- Lower protein
- Lower LDH
- Does not satisfy Light's exudative criteria

28. Exudative Effusion

Common causes include:

- Parapneumonic effusion
- Empyema
- Tuberculosis
- Malignancy

- Pulmonary embolism
- Pancreatitis
- Rheumatoid pleuritis
- Other inflammatory disorders

Typical findings:

- Increased protein
- Increased LDH
- Increased cells
- May be cloudy/turbid
- One or more Light's criteria positive

29. Light's Criteria and Heart Failure

Patients with heart failure, especially after diuretic therapy, may occasionally be classified as exudative by Light's criteria despite a cardiac mechanism.

In this situation additional parameters may be useful.

Serum–Pleural Albumin Gradient

Serum Albumin – Pleural Fluid Albumin

A gradient:

>1.2 g/dL

may support a transudative/cardiac mechanism.

Another supportive parameter is:

- Serum Protein – Pleural Protein >3.1 g/dL
- Clinical and cardiac correlation is necessary. (AAFP)

30. Pleural Fluid Glucose

Report:

Pleural Glucose: _____ mg/dL

Low pleural glucose may occur in:

- Pleural infection
- Empyema
- Rheumatoid pleuritis
- Malignancy
- Tuberculosis
- Esophageal rupture

Very low glucose should be interpreted together with pleural pH, LDH and microbiological findings.

31. Pleural Fluid pH

Pleural fluid pH is particularly useful when evaluating **parapneumonic effusion and suspected pleural infection**.

pH ≤7.20

Indicates a high probability of complicated parapneumonic effusion/pleural infection in the appropriate clinical setting and usually requires urgent clinical assessment for drainage. (Thorax)

pH >7.20 but <7.40

Represents an intermediate-risk range. Pleural LDH, glucose, imaging and clinical findings become particularly important.

pH ≥7.40

Suggests a lower risk of complicated parapneumonic infection in the appropriate setting. (Thorax)

32. Complicated Parapneumonic Effusion

Laboratory findings suggesting complicated pleural infection may include:

1. Purulent appearance
2. Positive Gram stain
3. Positive bacterial culture
4. $\text{pH} \leq 7.20$
5. Low glucose
6. Markedly increased LDH
7. Neutrophil predominance

Current BTS guidance considers a pleural $\text{pH} \leq 7.2$ high risk for complicated parapneumonic effusion or pleural infection. For $\text{pH} > 7.2$ and < 7.4 , $\text{LDH} > 900 \text{ U/L}$ together with other clinical features increases concern. (Thorax)

33. Empyema

Empyema is diagnosed when frank pus is present in the pleural space.

Laboratory appearance may be:

- Thick
- Turbid
- Purulent
- Yellow-green
- Foul smelling

Associated findings may include:

- Very high neutrophil count
- High LDH
- Low glucose
- Low pH
- Positive Gram stain
- Positive culture

Frankly purulent pleural fluid is itself diagnostic of pleural infection; pH measurement is not necessary to establish that diagnosis. (AAFP)

34. Gram Stain

Procedure

1. Mix specimen.
2. Centrifuge an appropriate quantity where indicated.
3. Prepare smear from sediment.
4. Air dry.
5. Fix.
6. Perform Gram stain.
7. Examine using oil immersion.

Report:

- No organisms seen
- or
- Gram-positive cocci seen
- Gram-negative bacilli seen etc.

Also report inflammatory response if relevant.

Example: **Numerous polymorphonuclear leukocytes seen with Gram-positive cocci.**

35. Culture

For suspected infection, investigate:

- Aerobic culture
- Anaerobic culture

When possible, submit:

- Plain sterile specimen
- Aerobic blood culture bottle
- Anaerobic blood culture bottle

BTS guidance recommends pleural fluid in both plain containers and blood-culture bottles when pleural infection is suspected. (Thorax)

36. Tuberculous Pleural Effusion

Common laboratory pattern includes:

- Exudative effusion
- Increased protein
- Lymphocyte predominance
- Elevated ADA in many cases
- AFB smear may be negative
- Mycobacterial culture may be positive
- Molecular testing may assist diagnosis

37. Pleural Fluid ADA

Adenosine deaminase may be useful in suspected tuberculous pleural effusion, particularly in settings with a high prevalence of tuberculosis.

A commonly used interpretive level is around:

ADA ≥ 40 U/L

which may support pleural tuberculosis in a lymphocyte-predominant exudate.

However:

1. Cutoffs vary with method and population.
2. Empyema and some other conditions can increase ADA.
3. A low ADA may be useful in making TB less likely.
4. ADA alone should never be used as definitive evidence of tuberculosis.

Current BTS guidance supports pleural ADA and/or interferon-gamma as useful tests in appropriate high-prevalence populations, while tissue sampling for microbiological confirmation remains important. (Thorax)

38. AFB / Tuberculosis Testing

When TB is suspected, consider:

- AFB smear
- Mycobacterial culture
- Molecular testing
- ADA
- Pleural biopsy where clinically indicated

A negative pleural-fluid AFB smear does not exclude tuberculosis.

39. Malignant Pleural Effusion

Pleural fluid may show:

- Bloody or serosanguineous appearance
- Exudative characteristics
- Lymphocyte predominance
- Elevated LDH
- Reduced glucose or pH in advanced disease
- Malignant cells on cytology

40. Cytological Examination

Cytology should be performed when malignancy is suspected.

Procedure

- Mix specimen.
- Centrifuge/cytocentrifuge.
- Prepare direct smear.
- Prepare cell block where available.
- Stain appropriately.
- Examine for:
 - Mesothelial cells
 - Macrophages
 - Inflammatory cells
 - Atypical cells
 - Malignant cells

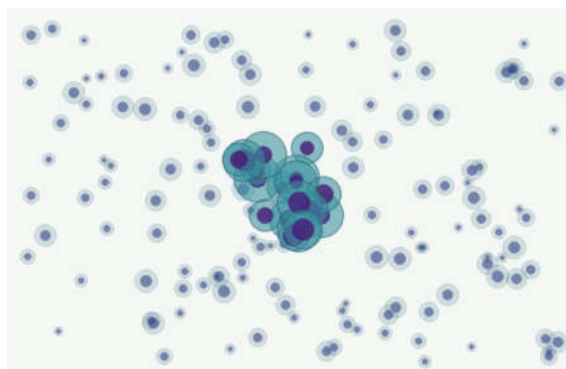
Thorax• Current BTS guidance recommends direct smear plus cell-block preparation and preferably 25–50 mL for initial cytology when available. (Thorax)

41. Cytology Reporting

Possible report terminology:

1. Negative for malignant cells
2. Reactive mesothelial cells present
3. Atypical cells present
4. Suspicious for malignancy
5. Malignant cells identified

A negative pleural-fluid cytology result does not exclude malignancy and may require further clinical investigation or pleural biopsy.



Pleural fluid cytology: clustered atypical / malignant-appearing cells (illustrative).

Illustrative microscopy reference — for visual correlation only; not a patient-specific result.

42. Chylothorax

Consider chylothorax when pleural fluid is:

- Milky
- White
- Opalescent

However, not all chylous effusions appear milky.

Measure:

- Pleural triglyceride
- Pleural cholesterol
- Chylomicrons where required

43. Chylothorax Interpretation

Pleural Triglyceride >110 mg/dL

with low cholesterol supports **chylothorax**.

Pleural Cholesterol <200 mg/dL

is commonly seen in chylothorax.

The definitive finding is demonstration of **chylomicrons**.

When triglyceride is approximately 50–110 mg/dL and clinical suspicion remains high, lipoprotein analysis for chylomicrons may be useful. (PubMed Central (PMC))

44. Causes of Chylothorax

Common causes include:

- Thoracic duct injury
- Thoracic surgery
- Trauma
- Lymphoma
- Other malignancy
- Lymphatic disorders

45. Pseudochylothorax

Pseudochylothorax may also produce milky fluid.

It usually occurs in long-standing pleural effusions.

Associated conditions include:

- Tuberculosis
- Rheumatoid pleuritis
- Chronic empyema

Typical laboratory features include:

- Increased pleural cholesterol
- Lower triglyceride
- Cholesterol crystals may be present

46. Hemothorax

When pleural fluid is grossly bloody, measure:

Pleural Fluid Hematocrit

and compare with peripheral blood hematocrit.

Calculation

Pleural Hct / Peripheral Blood Hct × 100

A pleural-fluid hematocrit **>50% of peripheral blood hematocrit** supports the diagnosis of hemothorax. (AAFP)

47. Example Hemothorax Calculation

Pleural Hct = 20%

Peripheral Hct = 35%

$20 \div 35 \times 100$

= 57%

Interpretation

Consistent with hemothorax

48. Pleural Fluid Amylase

Pleural amylase may be requested when considering:

- Pancreatitis
- Pancreatic pseudocyst
- Pancreatico-pleural fistula
- Esophageal rupture
- Some malignancies

Markedly increased pleural amylase should be interpreted together with serum amylase/lipase and clinical findings. (AAFP)

49. General Interpretation Table

Table 3.3. General Interpretation of Pleural Fluid

Condition	Light's Classification	Typical Cell Pattern	Additional Findings
Heart failure	Usually transudate	Low cells	Cardiac findings
Cirrhosis	Usually transudate	Low cells	Hepatic hydrothorax
Parapneumonic effusion	Exudate	Neutrophils	↓ pH, ↓ glucose, ↑ LDH
Empyema	Exudate	Neutrophils	Pus, very low pH/glucose
Tuberculosis	Exudate	Lymphocytes	↑ ADA commonly
Malignancy	Exudate	Often lymphocytes	Cytology may be positive
Pulmonary embolism	Often exudate	Variable	May be bloody
Pancreatic disease	Exudate	Variable	↑ Amylase
Rheumatoid pleuritis	Exudate	Variable	Very low glucose possible
Chylothorax	Usually exudate	Often lymphocytes	TG >110 mg/dL
Hemothorax	Variable	RBC-rich	Pleural Hct >50% of blood Hct

50. Quality Control

Cell Count

1. Mix specimen thoroughly but gently.
2. Check chamber cleanliness.
3. Avoid air bubbles.
4. Avoid chamber overfilling.
5. Perform duplicate counts according to laboratory policy.
6. Verify flagged automated results microscopically.

7. Use only analyzers validated for body fluids.

Differential Count

1. Prepare satisfactory smears.
2. Use appropriate stain.
3. Count adequate cells.
4. Refer atypical cells for experienced review.

Biochemistry

1. Perform routine internal QC.
2. Ensure assays used for body fluids are validated/verified according to laboratory policy.
3. Use appropriate calibration.
4. Record serum and fluid samples correctly to prevent calculation errors.

pH

1. Use a blood gas analyzer.
2. Prevent air contamination.
3. Avoid local anesthetic contamination.
4. Avoid excessive liquid heparin.
5. Analyze promptly.

Microbiology

Use appropriate routine bacteriological QC procedures.

51. Limitations

1. Light's criteria may classify some diuretic-treated cardiac effusions as exudative.
2. Cell differential alone is not diagnostic of a specific disease.
3. Negative Gram stain does not exclude pleural infection.
4. Negative bacterial culture does not exclude infection.
5. Antibiotics given before collection reduce culture yield.
6. Negative AFB smear does not exclude pleural TB.
7. Elevated ADA is not specific for tuberculosis.
8. Negative cytology does not exclude malignancy.
9. Pleural pH is highly affected by specimen collection and handling.
10. Appearance alone cannot distinguish chylothorax from pseudochylothorax or empyema.
11. Light's criteria require correctly paired pleural and serum values.

52. Safety Precautions

Treat all pleural fluid specimens as potentially infectious.

Use:

- Gloves
- Laboratory coat/gown
- Eye protection where splash is possible
- Appropriate biological safety cabinet for aerosol-generating procedures

Dispose of contaminated specimens and materials according to biomedical waste policy.

53. READY-TO-USE PLEURAL FLUID REPORT FORMAT

LABORATORY NAME: _____			
PLEURAL FLUID ANALYSIS REPORT			
PATIENT / SPECIMEN INFORMATION			
Patient Name	_____	Age/Sex	_____
Patient ID	_____	Ward / OPD	_____
Clinical Diagnosis	_____	Specimen	Pleural Fluid
Date/Time Collected	_____	Date/Time Received	_____
Side	☐ Right ☐ Left ☐ Not specified	Other	_____
PHYSICAL EXAMINATION			
Parameter	Result	Unit	
Volume Received	_____	mL	
Color	_____		
Appearance / Clarity	_____		
Clot	_____		
CELL COUNT			
Parameter	Result	Unit	
Total Nucleated Cells / WBC	_____	cells/ μ L	
RBC	_____	cells/ μ L	
DIFFERENTIAL COUNT			
Cell Type	Result		
Neutrophils	_____ %		
Lymphocytes	_____ %		
Monocytes / Macrophages	_____ %		
Eosinophils	_____ %		
Mesothelial Cells	_____ %		
Others	_____ %		
BIOCHEMICAL EXAMINATION			
Test	Pleural Fluid	Serum	Unit
Total Protein	_____	_____	g/dL
LDH	_____	_____	U/L
Albumin	_____	_____	g/dL
Glucose	_____	—	mg/dL
pH	_____	—	—
Amylase	_____	_____	U/L
Triglyceride	_____	—	mg/dL
Cholesterol	_____	—	mg/dL
ADA	_____	—	U/L
Other	_____	_____	_____
Serum LDH Upper Reference Limit	_____ U/L		
LIGHT'S CRITERIA CALCULATION			
1. Pleural Protein / Serum Protein	_____ $\div$ _____ = _____ ☐ >0.5 Positive ☐ $\leq$ 0.5 Negative		

LABORATORY NAME: _____			
2. Pleural LDH / Serum LDH	_____ ÷ _____ = _____ ☐ >0.6 Positive ☐ ≤0.6 Negative		
3. Pleural LDH vs ⅔ Serum LDH URL	Serum LDH URL _____ U/L ⅔ URL _____ U/L Pleural LDH _____ U/L ☐ Pleural LDH >⅔ URL — Positive ☐ Pleural LDH ≤⅔ URL — Negative		
LIGHT'S CLASSIFICATION			
Classification	☐ EXUDATIVE ☐ TRANSUDATIVE (One or more positive criteria = exudative)		
MICROBIOLOGY			
Gram Stain	_____		
Aerobic Culture	_____		
Anaerobic Culture	_____		
AFB Stain	_____		
Mycobacterial Culture / Molecular Test	_____		
CYTOLOGY			
Result	☐ Negative for malignant cells ☐ Reactive mesothelial cells ☐ Atypical cells ☐ Suspicious for malignancy ☐ Malignant cells identified ☐ Other: _____		
SPECIAL TEST INTERPRETATION			
Pleural ADA	Result: _____ U/L Comment: _____		
Triglyceride	Result: _____ mg/dL ☐ Chylothorax pattern ☐ Not suggestive ☐ Equivocal		
Pleural Hematocrit	Pleural Hct _____ % Peripheral Hct _____ % Ratio _____ % ☐ Consistent with hemothorax ☐ Not consistent with hemothorax		
INTERPRETATION / COMMENTS			
Reported By	Checked By	Authorized By	Date/Time
_____	_____	_____	_____

54. Suggested Standard Laboratory Comments

Exudative Effusion

“Pleural fluid fulfills one or more Light's criteria and is classified as an exudative effusion. Clinical correlation and evaluation for infective, malignant, tuberculous, inflammatory or other causes are recommended.”

Transudative Effusion

“Pleural fluid does not fulfill Light's criteria for an exudate and is classified as a transudative effusion. Correlation with cardiac, hepatic, renal and other systemic causes is recommended.”

Suspected Pleural Infection

“Pleural fluid shows neutrophilic inflammation. Correlation with pH, glucose, LDH, Gram stain and culture is recommended to assess for complicated parapneumonic effusion/pleural infection.”

Low Pleural pH

“Pleural fluid pH ≤7.20 in the appropriate clinical setting is associated with a high risk of complicated parapneumonic effusion/pleural infection. Urgent clinical assessment is recommended.”

Empyema

“Grossly purulent pleural fluid is consistent with empyema/pleural infection. Urgent microbiological and clinical correlation is advised.”

Tuberculosis Pattern

“Lymphocyte-predominant exudative pleural effusion with elevated ADA may support tuberculous pleuritis in the appropriate clinical setting. Correlation with mycobacterial culture, molecular testing and/or tissue diagnosis is recommended.”

Malignancy

“Malignant cells are identified in the pleural fluid. Correlation with cytomorphology, immunocytochemistry/histopathology and clinical findings is recommended.”

Negative Cytology

“No malignant cells identified in the examined specimen. A negative pleural-fluid cytology does not exclude malignancy.”

Chylothorax

“Elevated pleural triglyceride with an appropriate lipid profile is consistent with chylothorax. Demonstration of chylomicrons may be used for confirmation when required.”

Hemothorax

“Pleural-fluid hematocrit exceeds 50% of peripheral blood hematocrit, supporting a diagnosis of hemothorax in the appropriate clinical setting.”

55. Critical / Urgent Result Communication

The laboratory should establish locally approved critical-result criteria.

Possible findings requiring urgent communication include:

- Frank pus/empyema
- Positive pleural-fluid Gram stain
- Positive culture from pleural fluid
- Pleural pH ≤ 7.20 in suspected pleural infection
- Findings highly suggestive of complicated parapneumonic effusion
- Significant hemothorax
- Unexpected malignant cells according to institutional policy

Document:

Critical Result: _____ **Communicated To:** _____

Date: _____ **Time:** _____

Communicated By: _____

56. Practical Laboratory Workflow

Figure 3.1. Pleural Fluid laboratory workflow

PLEURAL FLUID RECEIVED



Verify patient identification, side and specimen



Record volume, color and appearance



Distribute specimen

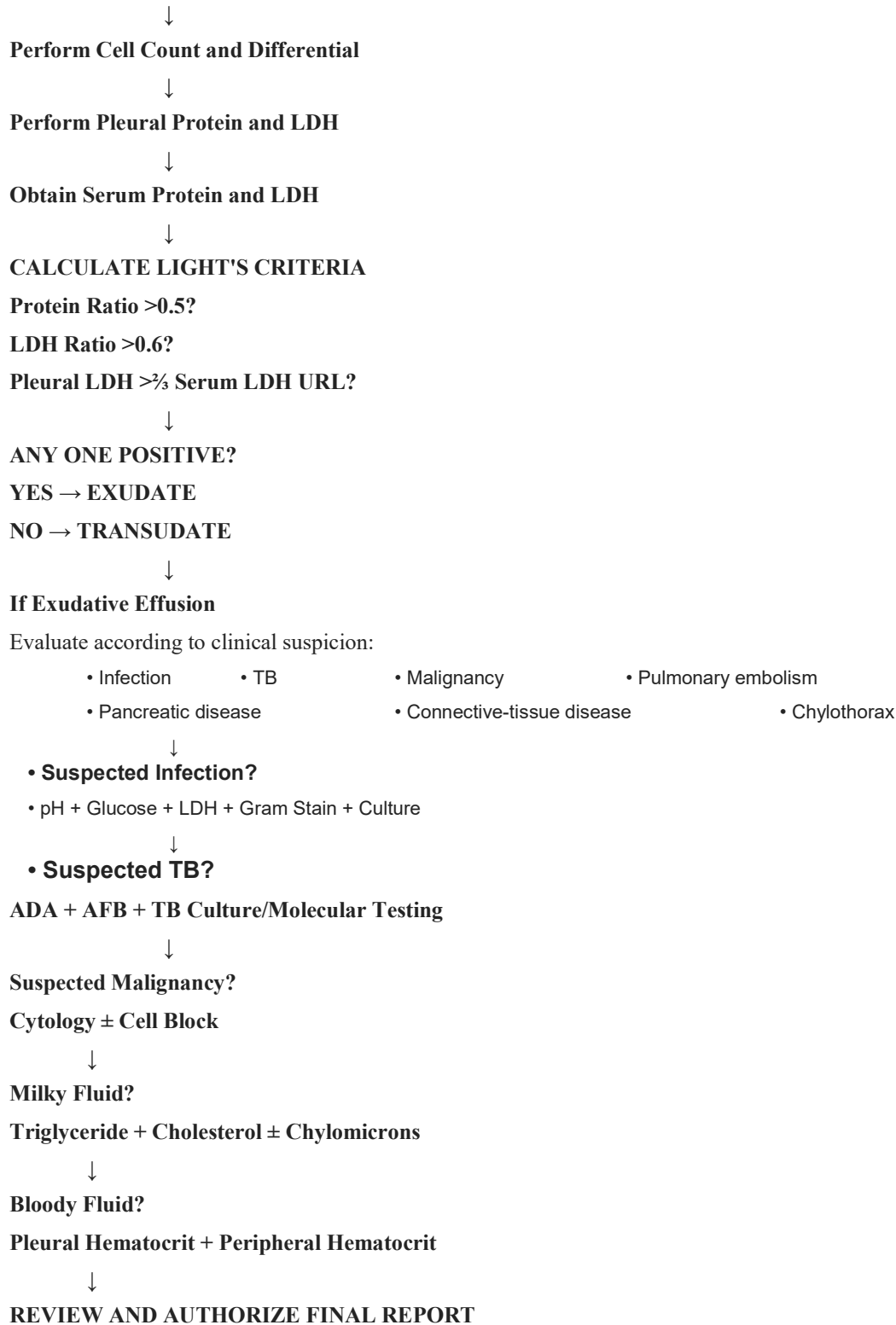
EDTA → Cell Count & Differential

Plain Tube → Protein / LDH / Glucose / Special Chemistry

Blood Gas Syringe → pH

Sterile Container + Blood Culture Bottles → Microbiology

Cytology Container → Cytology



57. Quick Interpretation Guide

Protein Ratio >0.5

→ Exudate Criterion Positive

LDH Ratio >0.6

→ Exudate Criterion Positive

Pleural LDH >2/3 Serum LDH Upper Limit

→ Exudate Criterion Positive

Any One Light's Criterion Positive

→ **EXUDATIVE EFFUSION**

Pleural pH ≤ 7.20 + Clinical Infection

→ **High Risk of Complicated Parapneumonic Effusion**

Frank Pus

→ **EMPYEMA**

Neutrophil Predominance

→ Consider **Acute Infection / Parapneumonic Effusion**

Lymphocyte Predominance

→ Consider **TB / Malignancy / Other Chronic Effusion**

Lymphocytes + Exudate + Increased ADA

→ Consider **Tuberculous Pleural Effusion**

Malignant Cells

→ **Malignant Pleural Effusion**

Triglyceride >110 mg/dL

→ Consider **Chylothorax**

Pleural Hct $>50\%$ Peripheral Blood Hct

→ **Hemothorax**

Increased Pleural Amylase

→ Consider **Pancreatic Disease / Esophageal Rupture / Selected Malignancies**

58. Key Laboratory Points

1. Always obtain paired **serum protein and LDH** when Light's criteria are required.
2. **One positive Light's criterion is sufficient to classify pleural fluid as exudative.**
3. Pleural-fluid pH should be measured using a **blood gas analyzer**, not a routine pH strip.
4. Local anesthetic contamination can falsely lower pH.
5. Residual air and delay can alter pH results.
6. **Frank pus = empyema**, regardless of pH.
7. A negative Gram stain does not exclude pleural infection.
8. Negative cytology does not exclude malignancy.
9. Lymphocyte predominance plus increased ADA supports TB but is not by itself diagnostic.
10. Milky fluid should not automatically be reported as chylothorax; triglyceride/cholesterol testing is required.
11. Grossly bloody fluid should be evaluated with pleural and peripheral hematocrit when hemothorax is suspected.
12. All results must be interpreted together with clinical and radiological findings.

59. Conclusion

Pleural fluid analysis is an important component in the diagnosis of pleural effusion.

The most useful initial laboratory investigations are:

Physical Examination + Cell Count + Differential Count + Protein + LDH + Glucose + Appropriate Microbiology
with paired:

Serum Protein + Serum LDH

for calculation of:

LIGHT'S CRITERIA

The fundamental laboratory classification is:

Transudative vs Exudative Pleural Effusion

For suspected infection, the additional key parameters are:

Pleural pH + Glucose + LDH + Gram Stain + Culture

For specific causes, additional investigations include:

- ADA/AFB/TB culture → Tuberculosis
- Cytology → Malignancy
- Triglyceride/cholesterol/chylomicrons → Chylothorax
- Hematocrit → Hemothorax
- Amylase → Pancreatic disease/esophageal rupture

A systematic approach integrating physical, microscopic, biochemical, microbiological and cytological findings provides the most useful diagnostic information from pleural fluid analysis.

Chapter 4. Cerebrospinal Fluid (CSF) Analysis

Laboratory Procedure, Interpretation and Reporting Manual

1. Introduction

Cerebrospinal fluid (CSF) is a clear fluid surrounding the brain and spinal cord. Laboratory examination of CSF is essential in the diagnosis of:

- Bacterial meningitis
- Viral meningitis
- Tuberculous meningitis
- Fungal meningitis
- Subarachnoid hemorrhage
- Central nervous system malignancy
- Multiple sclerosis and other neurological disorders
- Guillain–Barré syndrome

CSF analysis usually includes physical examination, cell count, differential count, glucose, protein, microbiological examination and selected specialized investigations.

2. Purpose

To establish a standardized laboratory procedure for:

- CSF specimen reception and handling
- Physical examination
- WBC and RBC counting
- Differential leukocyte count
- Protein and glucose measurement
- CSF/serum glucose ratio
- Gram stain and culture
- TB and fungal investigation
- Cytology
- Specialized investigations
- Interpretation and reporting

3. Specimen

Specimen: Cerebrospinal Fluid

Usually obtained by:

- Lumbar puncture
- Ventricular drainage
- Cisternal puncture in selected settings
- Shunt/reservoir sampling

Opening pressure should be measured by the clinician during lumbar puncture where clinically appropriate.

Normal adult opening pressure is generally about **60–250 mm H₂O**, but age and clinical circumstances must be considered. (AAFP)

4. Specimen Collection

CSF is usually collected into sequential sterile tubes.

A practical allocation may be:

Table 4.1. CSF Specimen Collection

Tube	Suggested Use
Tube 1	Chemistry / serology
Tube 2	Microbiology
Tube 3	Cell count and differential
Additional tube	Cytology / molecular / special tests

Tube allocation may vary according to institutional policy and clinical indication.

When only a small amount of CSF is available, testing should be prioritized according to clinical urgency, especially microbiology and cell count when meningitis is suspected.

5. Specimen Transport

CSF is a **priority specimen** and should be transported immediately.

For suspected bacterial meningitis:

1. Do not unnecessarily delay processing.
2. Perform cell count, Gram stain and culture promptly.
3. Do not refrigerate a specimen intended for bacterial culture unless specifically required by the organism/testing protocol.

For molecular, viral, fungal or TB investigations, follow assay-specific storage requirements.

6. Specimen Rejection / Qualification

Because CSF is difficult and potentially hazardous to recollect, specimens should rarely be rejected automatically.

Potential problems include:

- Unlabeled specimen
- Incorrect identification
- Leaking container
- Gross external contamination
- Inappropriate container
- Excessive delay
- Insufficient quantity

Contact the responsible clinician before rejecting an irreplaceable specimen whenever possible.

7. Required Equipment and Materials

1. PPE
2. Hemocytometer or validated automated body-fluid analyzer
3. Centrifuge
4. Cytocentrifuge where available
5. Microscope
6. Glass slides
7. Wright-Giemsa or equivalent stain
8. Gram stain reagents
9. AFB staining reagents where required
10. India ink or cryptococcal antigen system where appropriate
11. Culture media

12. Biochemistry analyzer
13. CSF protein reagent
14. Glucose reagent
15. Molecular testing system where available

8. Laboratory Examination

Routine CSF examination includes:

- A. Physical examination
- B. Cell count
- C. Differential count
- D. Biochemistry
- E. Microbiology
- F. Special investigations

9. Physical Examination

Observe the specimen before centrifugation.

Record:

- Volume received
- Color
- Clarity
- Presence of blood
- Clot/pellicle where present

10. Normal Appearance

Normal CSF is typically:

Clear and colorless

Abnormal appearances include:

Table 4.2. CSF Normal and Abnormal Appearance

Appearance	Possible Association
Cloudy/turbid	Increased WBC, RBC, microorganisms
Purulent	Severe bacterial meningitis
Bloody	Traumatic tap or intracranial/subarachnoid hemorrhage
Xanthochromic	Breakdown of blood products, markedly increased protein or other causes
Fibrin web/pellicle	May occur in TB meningitis

Appearance alone must not be used to establish diagnosis.

11. Xanthochromia

Xanthochromia is a yellow, orange or pink discoloration of the CSF supernatant.

It may occur due to:

- Subarachnoid hemorrhage
- RBC breakdown
- Markedly increased protein
- Hyperbilirubinemia

When subarachnoid hemorrhage is suspected, interpretation should consider clinical timing, imaging and whether the specimen represents a traumatic puncture.

12. Total WBC Count

Cell counting may be performed manually or using a validated automated body-fluid analyzer.

General adult reference

WBC <5 cells/ μ L is commonly used as the adult reference interval. (AAFP)

Neonatal and pediatric reference intervals differ and should be age-specific.

13. Manual WBC Count

Procedure:

- Mix CSF gently.
- Load the counting chamber.
- Allow cells to settle.
- Count nucleated cells.
- Dilute only when cell concentration is too high for accurate counting.
- Calculate cells/ μ L.

General Formula:

Cells/ μ L = Cells counted $\times$ Dilution factor $\times$ Chamber factor

- Follow the validated counting-area calculation used by the laboratory.

14. RBC Count

Report: **RBC:** _____ cells/ μ L

RBCs may be present due to:

- Traumatic lumbar puncture
- Subarachnoid hemorrhage
- Intracranial hemorrhage
- CNS trauma
- Hemorrhagic lesion

Serial reduction in RBC count from the first to later collection tubes may support traumatic contamination but is not by itself sufficient to exclude subarachnoid hemorrhage.

15. Traumatic Tap and WBC Correction

Peripheral blood introduced during traumatic lumbar puncture may falsely increase the CSF WBC count.

A traditional approximation is to subtract approximately **1 WBC for every 500–1,500 RBCs**, but a more individualized estimate can use peripheral blood counts. (AAFP)

Predicted contaminating WBC

Predicted CSF WBC = CSF RBC $\times$ Peripheral Blood WBC / Peripheral Blood RBC

Corrected CSF WBC

Corrected WBC = Measured CSF WBC – Predicted contaminating WBC

This correction is only supportive and must **not** be used to rule out meningitis when clinical suspicion is high.

16. Differential Cell Count

Prepare a cytospin or concentrated smear where possible.

Report: